

CARIBBEAN EXAMINATIONS COUNCIL

**Caribbean Secondary Education Certificate®
CSEC®**



Mathematics

Specimen Papers and Mark Schemes/Keys

Specimen Papers, Mark Schemes and Keys:

- Paper 01
- Paper 02
- Paper 032



TEST CODE **01234010**

SPECIMEN 2025

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

MATHEMATICS

Paper 01 – General Proficiency

1 hour 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. **A list of formulae is provided on pages 2 and 3 of this booklet.**
4. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
5. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The expression $2p + 3(p - q)$ written in its simplest form is

Sample Answer

- (A) $5p - q$
- (B) $5p - 3q$
- (C) $5p + 3q$
- (D) $5p(1 - q)$



The best answer to this item is “ $5p - 3q$ ”, so (B) has been shaded.

6. If you want to change your answer, erase it completely before you fill in your new choice.
7. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
8. You may do any rough work in this booklet.
9. Calculators and mathematical tables are NOT allowed for this paper.
10. **Diagrams in this booklet are NOT drawn to scale, unless otherwise stated.**

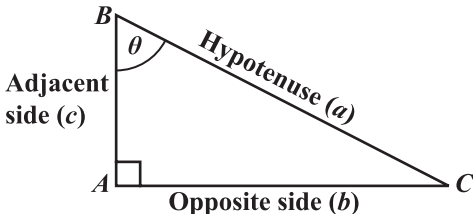
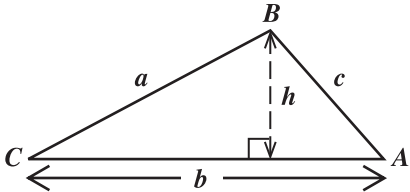
DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF FORMULAE

Volume of a prism $V = Ah$	A – area of a cross-section h – the perpendicular length
Volume of a cylinder $V = \pi r^2 h$	r – radius of the base h – the perpendicular height
Volume of a right circular cone/right pyramid $V = \frac{1}{3} Ah$	A – area of the base h – the perpendicular height
Curved surface area of a cone $CSA = \pi rl$	r – radius of the base l – the slant height of the cone
Surface area of a sphere $SA = 4\pi r^2$	r – radius of the sphere.
Volume of a sphere $V = \frac{4}{3} \pi r^3$	r – radius of the sphere
Circumference of a circle $C = 2\pi r$ $C = \pi d$	r – radius of the circle d – diameter of the circle
Arc length $S = \frac{\theta}{360} \times 2\pi r$	θ – the angle subtended by the arc, measured in degrees
Area of a circle $A = \pi r^2$	r – radius of the circle
Area of a sector $A = \frac{\theta}{360} \times \pi r^2$	θ – the angle of the sector, measured in degrees
Area of a trapezium $A = \frac{1}{2} (a + b) h$	a and b – the lengths of the parallel sides h – the perpendicular distance between the parallel sides
Simple interest $SI = \frac{P \times R \times T}{100}$	P – principal (initial amount) R – annual rate of interest T – time (in years)
Compound interest $A = P \left[1 + \frac{r}{100} \right]^n$	A – total amount after n years P – principal (initial amount) r – annual rate of interest n – number of years money is invested
Depreciation $A = P \left[1 - \frac{r}{100} \right]^n$	A – value of item after depreciation P – initial value of the item r – annual rate of depreciation n – number of years item depreciates

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LIST OF FORMULAE (continued)

Roots of quadratic equations If $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	a – the coefficient of x^2 b – the coefficient of x c – the constant term
Trigonometric ratios $\sin \theta = \frac{b}{a}$ $\cos \theta = \frac{c}{a}$ $\tan \theta = \frac{b}{c}$	 a – length of the hypotenuse b – length of the opposite side c – length of the adjacent side
Pythagoras' theorem $a^2 = b^2 + c^2$	
Area of a triangle $\text{Area of } \Delta = \frac{1}{2} bh$	 b – length of the base of the Δ h – the perpendicular height of the Δ
Area of $\Delta ABC = \frac{1}{2} ab \sin C$	a and b – the lengths of the adjacent sides of the Δ C – the included angle.
Area of $\Delta ABC =$ $\sqrt{s(s-a)(s-b)(s-c)}$	$s = \frac{a+b+c}{2}$ – the semi-perimeter of the Δ a, b and c – the sides of the Δ
Sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	a, b and c – the sides of the Δ A, B and C – the angles opposite the corresponding sides of the Δ
Cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$	a, b and c – the sides of the Δ A – the angle opposite Side a
Counting formula: union of 2 sets $n(A \cup B) =$ $n(A) + n(B) - n(A \cap B)$	A and B – two finite intersecting sets

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1. The number 3 857 written to 3 significant figures is

(A) 386
(B) 3 850
(C) 3 860
(D) 4 000

2. Expressed in scientific notation,

$$0.045 \times 10^{-3} \text{ is}$$

(A) 4.5×10^{-1}
(B) 4.5×10^{-4}
(C) 4.5×10^{-5}
(D) 4.5×10^{-6}

3. The value of $\frac{(5+2)^3}{5^2-2^2}$ in its simplest form is

(A) $\frac{8}{21}$
(B) $\frac{7}{3}$
(C) $\frac{7}{2}$
(D) $\frac{49}{3}$

4. There are 50 children in a youth club. Girls make up 60% of the club. Of the boys in the club, 25% of them play chess. How many boys play chess?

(A) 5
(B) 10
(C) 15
(D) 24

5. Tom bought a pen for \$60. He sold it to gain 20% on the cost price. The profit he gained was

(A) \$12
(B) \$40
(C) \$72
(D) \$80

6. How much simple interest is due on a loan of \$1 200 for two years, if the annual rate of interest is 5%?

(A) \$120
(B) \$132
(C) \$264
(D) \$330

Item 7 refers to the following chart which shows rates on fixed deposits for 2023 and 2024.

Rates on Fixed Deposits	
2023	7.8%
2024	7.5%

7. How much MORE interest did a fixed deposit of \$10 000 earn in 2023 than in 2024?

(A) \$ 0.30
(B) \$ 3.00
(C) \$30.00
(D) \$33.00

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8. The Water Authority charges \$10 per month for the rental of the meter, \$25 for the first 100 litres and \$1 for each additional 10 litres of water used.

Jonathan's bill for December was \$50. What is the LEAST amount of water he most likely used?

- (A) 250 litres
- (B) 260 litres
- (C) 300 litres
- (D) 350 litres

9. Which of the following sets has an infinite number of elements?

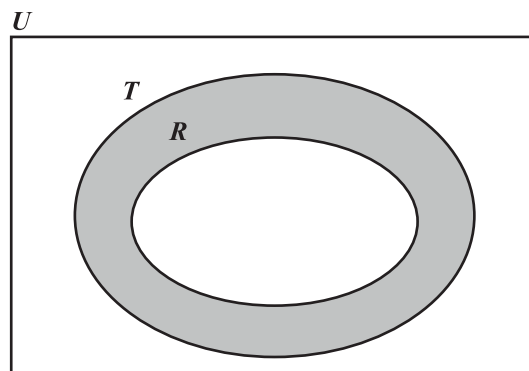
- (A) {Factors of 20}
- (B) {Multiples of 20}
- (C) {Prime numbers less than 20}
- (D) {Odd numbers between 10 and 20}

Items **10** and **11** refer to the following Venn diagram which represents the relationship between T and R for students in Form 5B, as defined below.

U = students in Form 5B

T = students who like thriller movies

R = students who like romance movies



10. The shaded area in the Venn diagram above represents students who like

- (A) romance movies
- (B) thriller but not romance movies
- (C) romance but not thriller movies
- (D) both romance and thriller movies

11. Of the 36 students in Form 5B, $n(T') = 7$. If the number of students in the shaded area in the Venn diagram is the same as $n(T')$, then the number of students in $T \cap R$ is

- (A) 7
- (B) 14
- (C) 22
- (D) 29

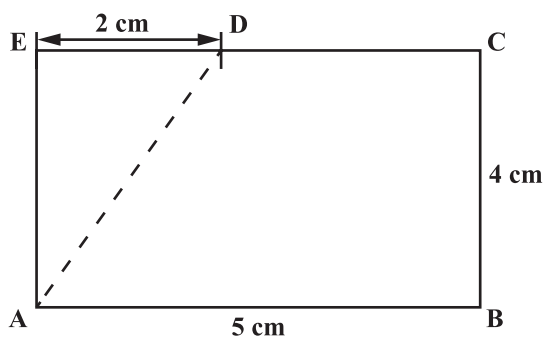
12. A man leaves home at 22:15 hours and reaches his destination at 04:00 hours on the following day, in the same time zone. How long did the journey take?

(A) $5\frac{1}{4}$ hours
 (B) $5\frac{3}{4}$ hours
 (C) 6 hours
 (D) $6\frac{1}{4}$ hours

13. How many litres of water would a container whose capacity is 36 cm^3 hold when filled?

(A) 0.036
 (B) 0.36
 (C) 36
 (D) 3 600

Item 14 refers to the following diagram of a rectangle, $ABCE$, which consists of a trapezium, $ABCD$, and a triangle, ADE .

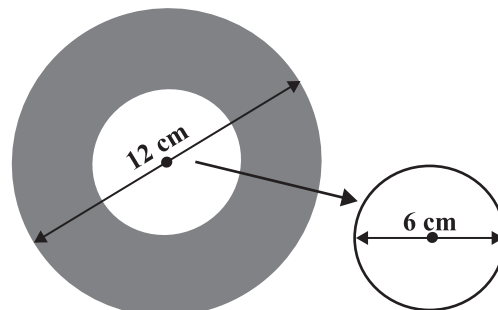


14. In the diagram above, the angles at B , C and E are right angles.

The area of the trapezium, $ABCD$, is

(A) 8 cm^2
 (B) 16 cm^2
 (C) 30 cm^2
 (D) 32 cm^2

15. A circular hole with a diameter of 6 cm is cut from a circular piece of card that has a diameter of 12 cm, as shown in the diagram below.



The area of the remaining card, in cm^2 , is

(A) 6π
 (B) 27π
 (C) 36π
 (D) 108π

16. Seven times a number p plus the number q may be written as

(A) $7pq$
 (B) $49pq$
 (C) $7p + q$
 (D) $7(p + q)$

17. If $2(x - 1) - 3x = 6$, then $x =$

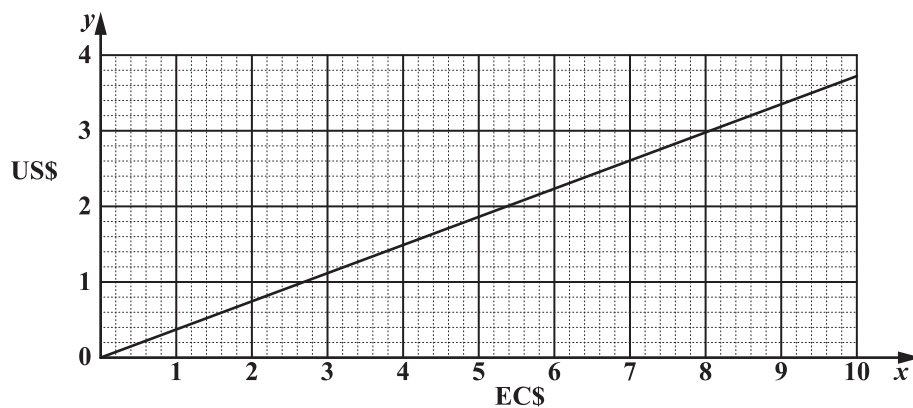
(A) -8
 (B) -4
 (C) 4
 (D) 8

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18. The result of $\frac{3x+1}{2} - \frac{x}{4}$ is

- (A) $\frac{2x+1}{4}$
- (B) $\frac{2x+2}{4}$
- (C) $\frac{5x+1}{4}$
- (D) $\frac{5x+2}{4}$

Items 19 and 20 refer to the following graph which shows the conversion rate with respect to United States dollars (US\$) and Eastern Caribbean dollars (EC\$).



19. Based on the graph, the coordinates of the x -intercept are

- (A) (0, 0)
- (B) (0, 1)
- (C) (1, 0)
- (D) (1, 1)

20. Using the graph, EC\$7 is equivalent to US\$

- (A) 2.00
- (B) 2.60
- (C) 3.00
- (D) 4.00

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Items **21–23** refer to the following table which shows the distribution of scores for 25 students who took a spelling test that was marked out of 10.

Score	4	5	6	8	9	10
Number of Students	5	3	5	3	4	5

- 21.** The number of students who scored AT MOST 8 marks on the test is
- (A) 3
(B) 9
(C) 12
(D) 16
- 22.** The median score of the distribution of scores for the 25 students is
- (A) 5
(B) 6
(C) 7
(D) 8
- 23.** What is the probability that a student chosen at random scores AT LEAST 5 marks on the test?
- (A) $\frac{1}{5}$
(B) $\frac{9}{25}$
(C) $\frac{16}{25}$
(D) $\frac{4}{5}$

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24. The mean of 10 numbers is 58. If one of the numbers is 40, what is the mean of the other 9?

(A) 18
(B) 60
(C) 68
(D) 540

25. The factorisation of $x^2 + 6x + 9$ is

(A) $(x - 3)^2$
(B) $(x + 3)^2$
(C) $(x - 3)(x + 3)$
(D) $(2x + 3)(x + 3)$

26. If $xy = yz - x$, then $x =$

(A) $\frac{yz}{y + 1}$
(B) $\frac{yz}{y - 1}$
(C) $\frac{-yz}{y + 1}$
(D) $\frac{y}{z(y + 1)}$

Item 27 refers to the following information.

The force, F , applied to an object is directly proportional to the extension, e , produced by that object. The incomplete table below shows the corresponding values of F and e .

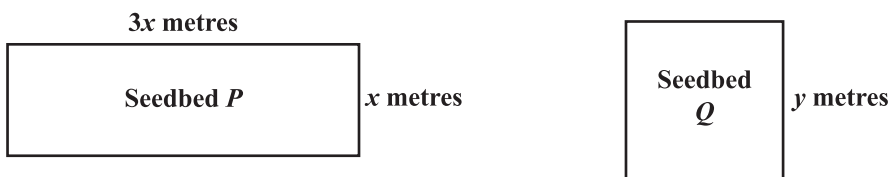
F	8	25	60
e	0.2	x	1.5

27. According to the table, the value of x is

(A) 0.375
(B) 0.625
(C) 1.225
(D) 1.875

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Item **28** refers to the following diagram which shows 2 seedbeds, P and Q . Seedbed P is a rectangle and Seedbed Q is a square with dimensions as shown in the diagram.



- 28.** If 60 m of mesh was used to enclose the 2 seedbeds, then a simplified expression for y in terms of x is

- (A) $y = 15 - 2x$
- (B) $y = 2x + 15$
- (C) $y = 8x - 60$
- (D) $y = 3x + 60$

-
- 29.** The equation of the line which passes through the point $(0, 2)$ and has a gradient of $\frac{1}{3}$ is

- (A) $y = 3x$
- (B) $y = \frac{1}{3}x$
- (C) $y = 3x + 2$
- (D) $y = \frac{1}{3}x + 2$

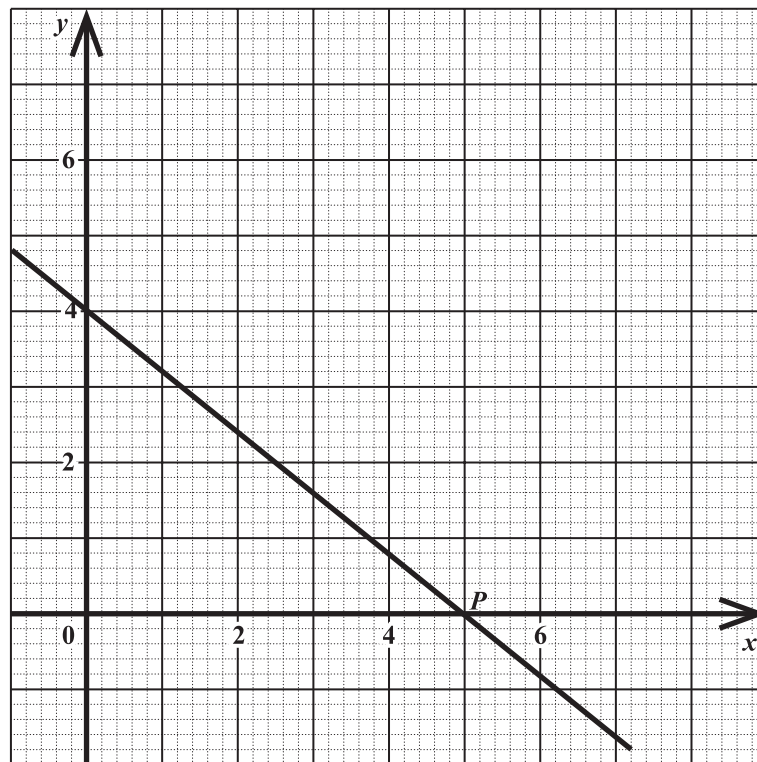
- 30.** If g is a function such that $g(x) = 2x + 1$, which of the following coordinates satisfy this function?

- (A) $(5, 2)$
- (B) $(13, 6)$
- (C) $(-3, -5)$
- (D) $(-6, 11)$

- 31.** If $f(x) = 2x - 3$ and $g(x) = 3x + 1$, then $fg(-2)$ is

- (A) -13
- (B) -7
- (C) 5
- (D) 20

Item 32 refers to the following diagram which shows a graph of a straight line intersecting the x and y axes.



32. The equation of the perpendicular line that passes through P is

(A) $y = -\frac{4}{5}x + 4$

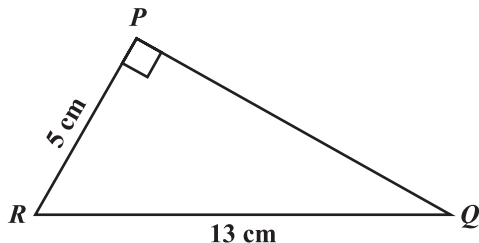
(B) $y = \frac{5}{4}x - \frac{25}{4}$

(C) $y = \frac{5}{4}x + \frac{25}{4}$

(D) $y = -\frac{4}{5}x - \frac{25}{4}$

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Items **33** and **34** refer to the following diagram which shows Triangle PQR in which angle $RPQ = 90^\circ$, $RP = 5$ cm and $RQ = 13$ cm.



33. The length of PQ , in cm, is

- (A) 7
- (B) 11
- (C) 12
- (D) 17

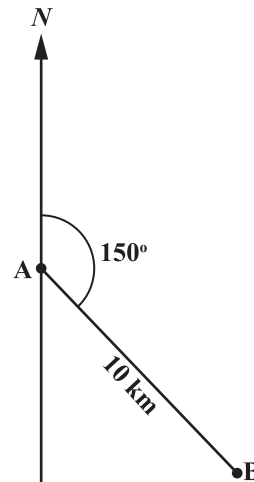
34. Which of the following trigonometric ratios

is/are equivalent to $\frac{5}{13}$?

- I. $\cos R$
- II. $\sin Q$
- III. $\tan R$

- (A) III only
- (B) I and II only
- (C) I and III only
- (D) II and III only

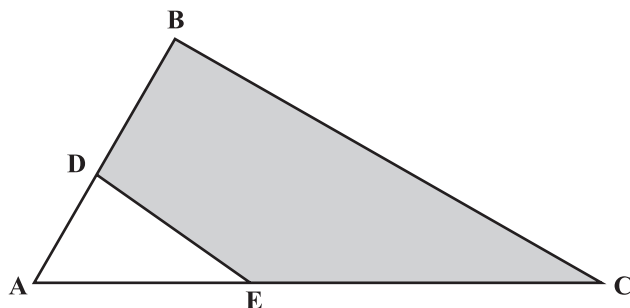
Item **35** refers to the following diagram which shows the path of a plane that travels from Point A on a bearing of 150° to Point B which is 10 km from A .



35. The distance of B due east of A is given by

- (A) $10 \tan 30^\circ$
- (B) $10 \cos 30^\circ$
- (C) $10 \sin 60^\circ$
- (D) $10 \cos 60^\circ$

Item **36** refers to the following diagram which shows that Triangle ABC is an enlargement of Triangle ADE such that $\frac{AD}{DB} = \frac{AE}{EC} = \frac{1}{2}$.



- 36.** If the area of $ABC = 36 \text{ cm}^2$, then the area of the shaded region, $DECB$, in cm^2 , is
- (A) 18
 - (B) 24
 - (C) 27
 - (D) 32

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Items 37 and 38 refer to the following matrix labelled R .

$$R = \begin{pmatrix} 2 & 5 & 9 \\ 0 & 1 & -1 \\ 6 & 3 & -3 \\ -4 & 2 & 8 \end{pmatrix}$$

37. The order of Matrix R is

- (A) 3×3
- (B) 3×4
- (C) 4×3
- (D) 4×4

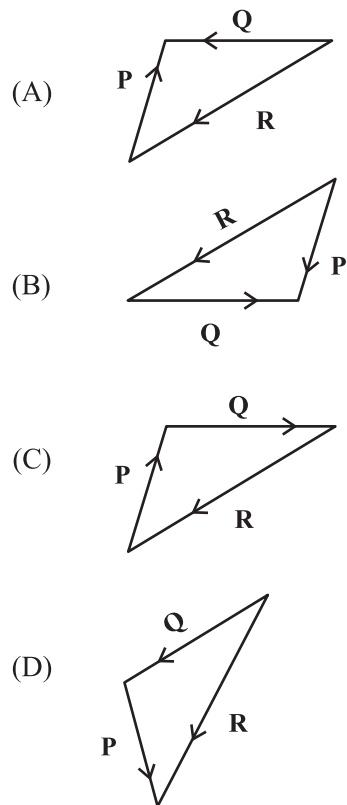
38. Matrix R is **pre-multiplied** by another matrix, B . If the order of the matrix product of the 2 matrices B and R is 3×3 , then the order of Matrix B is

- (A) 3×3
- (B) 3×4
- (C) 4×3
- (D) 4×4

39. Which of the following groups of quantities represents a set of scalar quantities?

- (A) Mass, time, volume
- (B) Force, frequency, acceleration
- (C) Displacement, speed, acceleration
- (D) Displacement, force, temperature

40. Which of the following diagrams represents the vector sum $\mathbf{R} = \mathbf{P} + \mathbf{Q}$?



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Items **41** and **42** refer to the following frequency distribution which shows the marks received by a group of young adults on a driving test.

Marks	11–20	21–30	31–40	41–50	51–60
Frequency	9	7	4	3	7

41. How many young adults scored GREATER THAN 40 marks on the test?

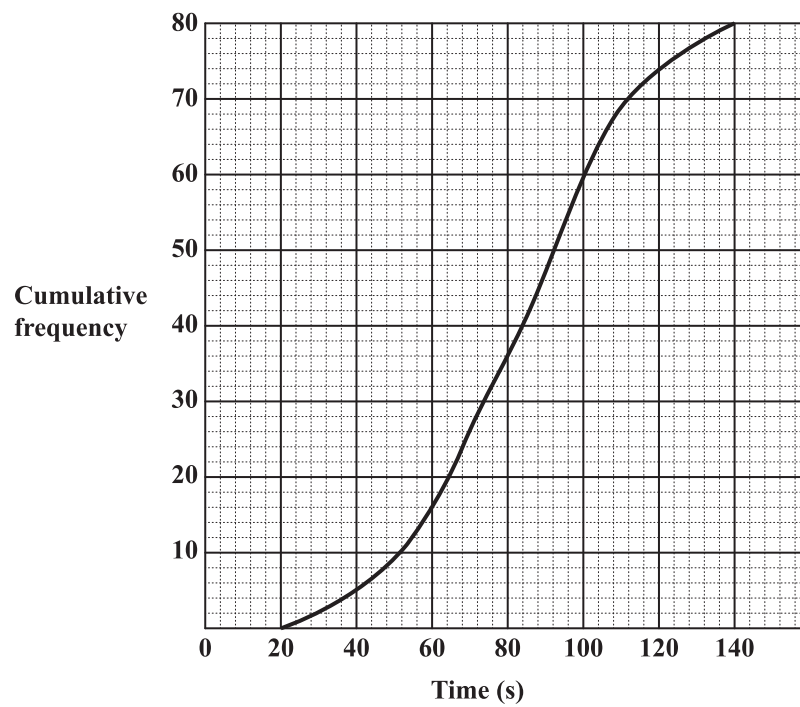
- (A) 7
- (B) 10
- (C) 14
- (D) 20

42. The lower limit of the median class is

- (A) 11
- (B) 21
- (C) 31
- (D) 41

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Item 43 refers to the following cumulative frequency curve for the time t , in seconds, taken by 80 students to complete a mathematical task.



43. According to the graph, the semi-interquartile range is

- (A) 18 s
- (B) 36 s
- (C) 50 s
- (D) 64 s

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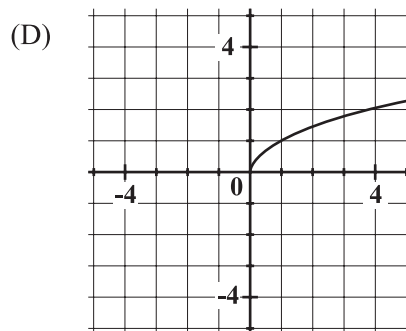
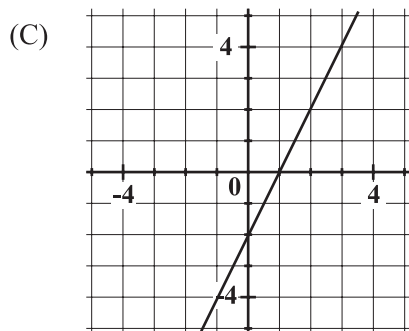
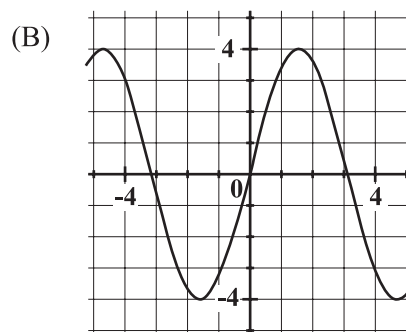
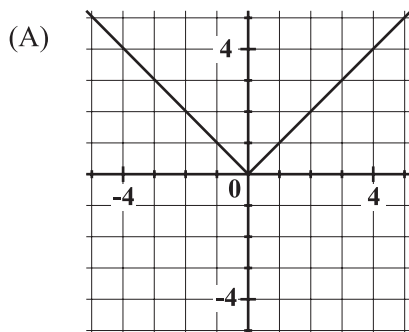
Item 44 refers to the following contingency table which shows the results of a survey of 10 families who were asked whether they have a cat or a dog or neither.

	A Dog	No Dog	Total
A Cat	1	2	3
No Cat	3	4	7
Total	4	6	10

44. A family was chosen at random from the 10 families surveyed and was found to have a dog. What is the probability that this family also owns a cat?

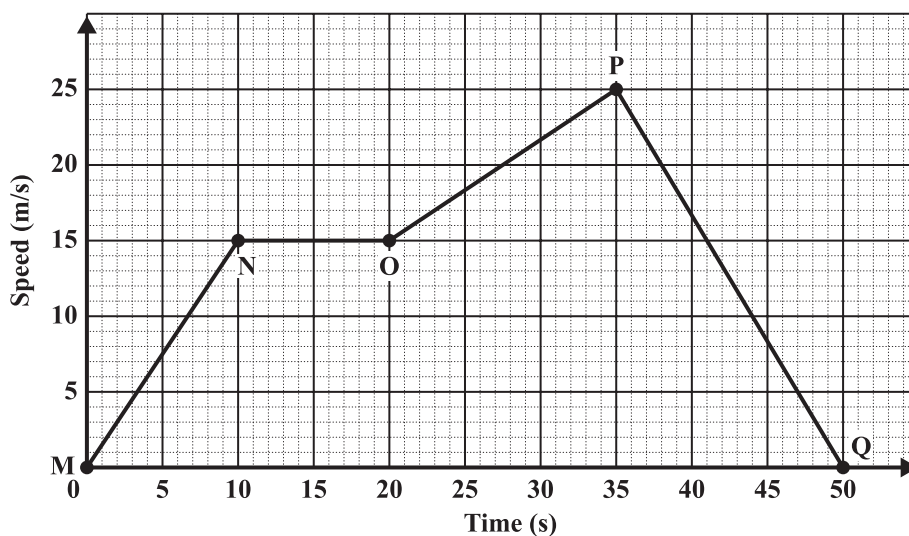
- (A) $\frac{1}{10}$
(B) $\frac{1}{4}$
(C) $\frac{3}{10}$
(D) $\frac{4}{10}$

45. In which of the following diagrams does the graph represent a linear function?



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Items 46 and 47 refer to the following diagram which shows a speed–time graph for the motion of an object along a straight line.



46. During which phase of its journey is the object travelling at constant speed?

- (A) *NO*
- (B) *OP*
- (C) *PQ*
- (D) *MN*

47. According to the graph, the acceleration of the object in the first 10 seconds is

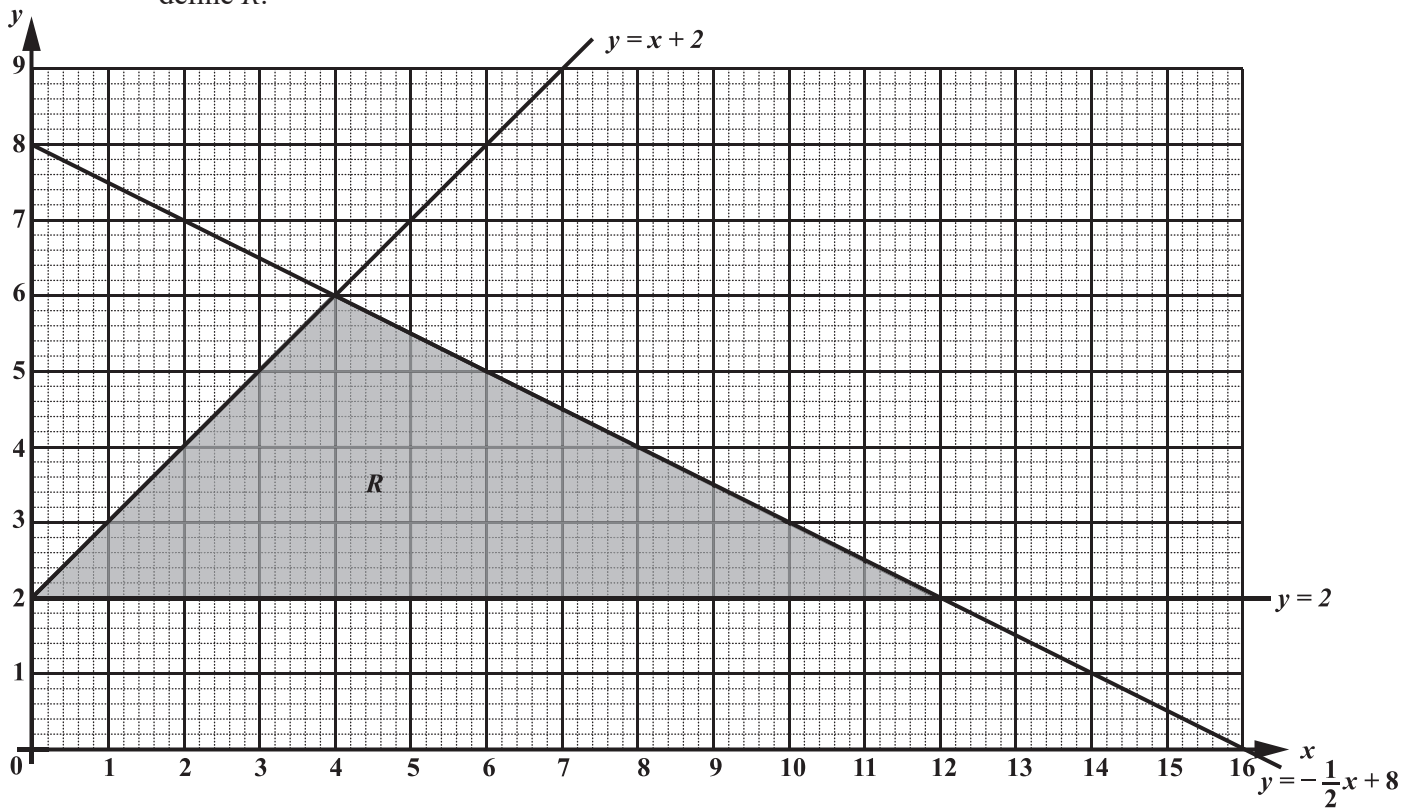
- (A) 0.67 ms^{-2}
- (B) 1.5 ms^{-2}
- (C) 15.0 ms^{-2}
- (D) 25.0 ms^{-2}

48. Which of the following line graphs represents the solution set for the inequality $x < 2x - 4 < 12$?

- (A)
- (B)
- (C)
- (D)

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Items 49 and 50 refer to the following diagram which shows the graph of 3 lines and a shaded region, R , which represents the common region for 3 inequalities associated with the lines that define R .



49. The system of inequalities that satisfies the region R is

- (A) $y \leq x + 2$
 $y \leq -\frac{1}{2}x + 8$
 $y \geq 2$
- (B) $y \geq x + 2$
 $y \geq -\frac{1}{2}x + 8$
 $y \geq 2$
- (C) $y \leq x + 2$
 $y \leq -\frac{1}{2}x + 8$
 $y \leq 2$
- (D) $y \geq x + 2$
 $y \geq -\frac{1}{2}x + 8$
 $y \leq 2$

50. In the diagram, the shaded region, R , represents the only possible number of toy phones (x) and toy trucks (y) that a manufacturer can make.

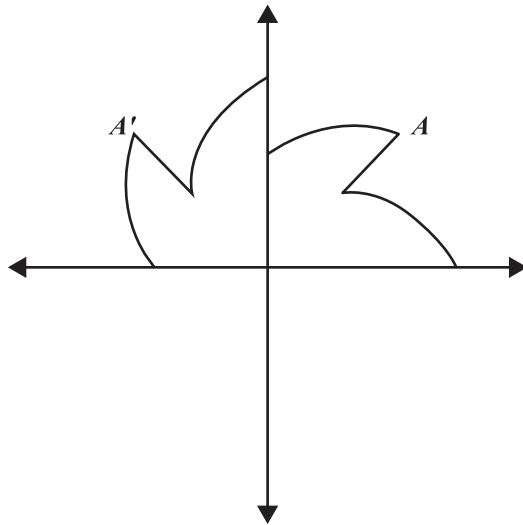
According to the graph, the number of phones and the number of trucks which will give the MAXIMUM profit is MOST likely

	x	y
(A)	0	2
(B)	8	3
(C)	4	7
(D)	12	2

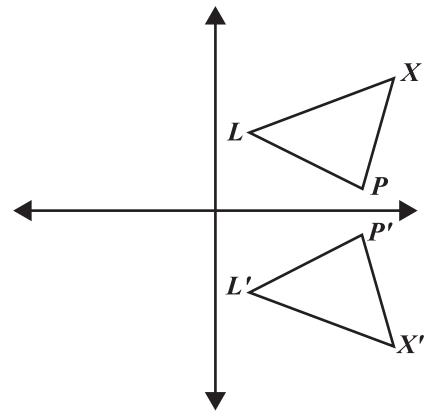
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51. Which of the following diagrams represents a reflection in a line?

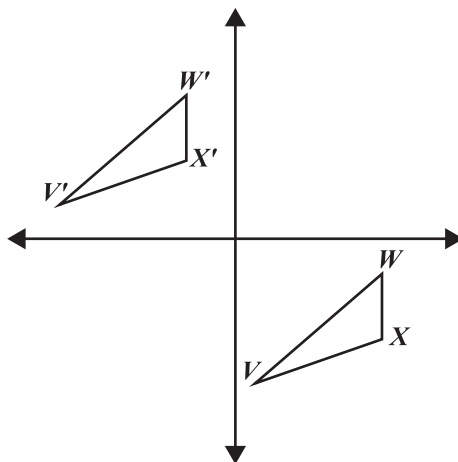
(A)



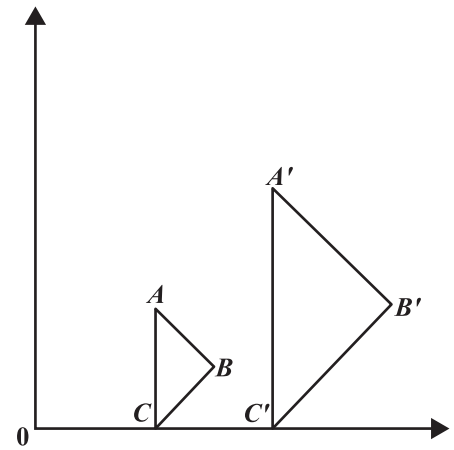
(B)



(C)



(D)

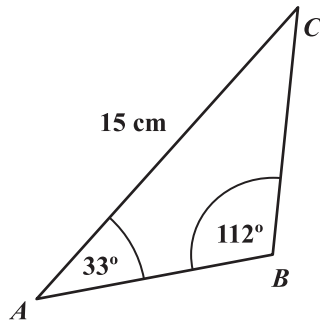


52. The point $P(2, -3)$ is rotated about the origin through an angle of 90° in an anti-clockwise direction.

What are the coordinates of the image of P ?

- (A) $(3, 2)$
- (B) $(2, 3)$
- (C) $(-3, 2)$
- (D) $(3, -2)$

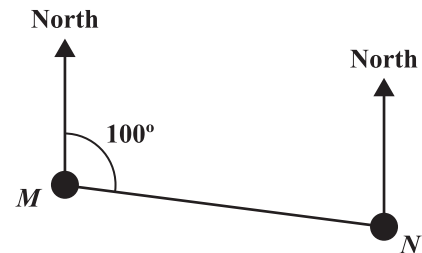
Item 53 refers to the following diagram which shows Triangle ABC where the length of AC is 15 cm, Angle $ABC = 112^\circ$ and Angle $BAC = 33^\circ$.



53. The length of BC is given by

- (A) $\frac{15 \sin 33^\circ}{\sin 112^\circ}$
- (B) $\frac{15 \sin 112^\circ}{\sin 33^\circ}$
- (C) $\sqrt{225 - 495 \cos 33^\circ}$
- (D) $\sqrt{495 - 225 \cos 112^\circ}$

Item 54 refers to the following diagram which shows the location of 2 points, M and N , on level ground.

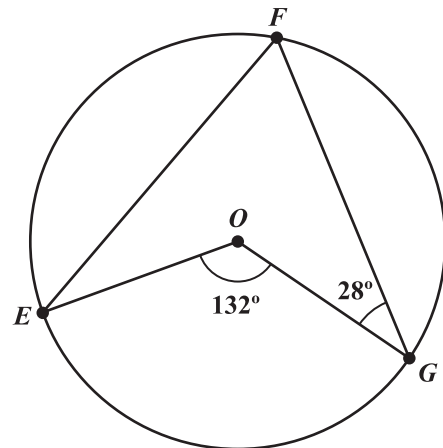


54. On the diagram, the bearing of N from M is 100° . What is the bearing of M from N ?

- (A) 80°
- (B) 120°
- (C) 260°
- (D) 280°

Item 55 refers to the following diagram which shows a circle with centre O and the points E , F and G on its circumference.

Angle $EOG = 132^\circ$.
Angle $FGO = 28^\circ$.

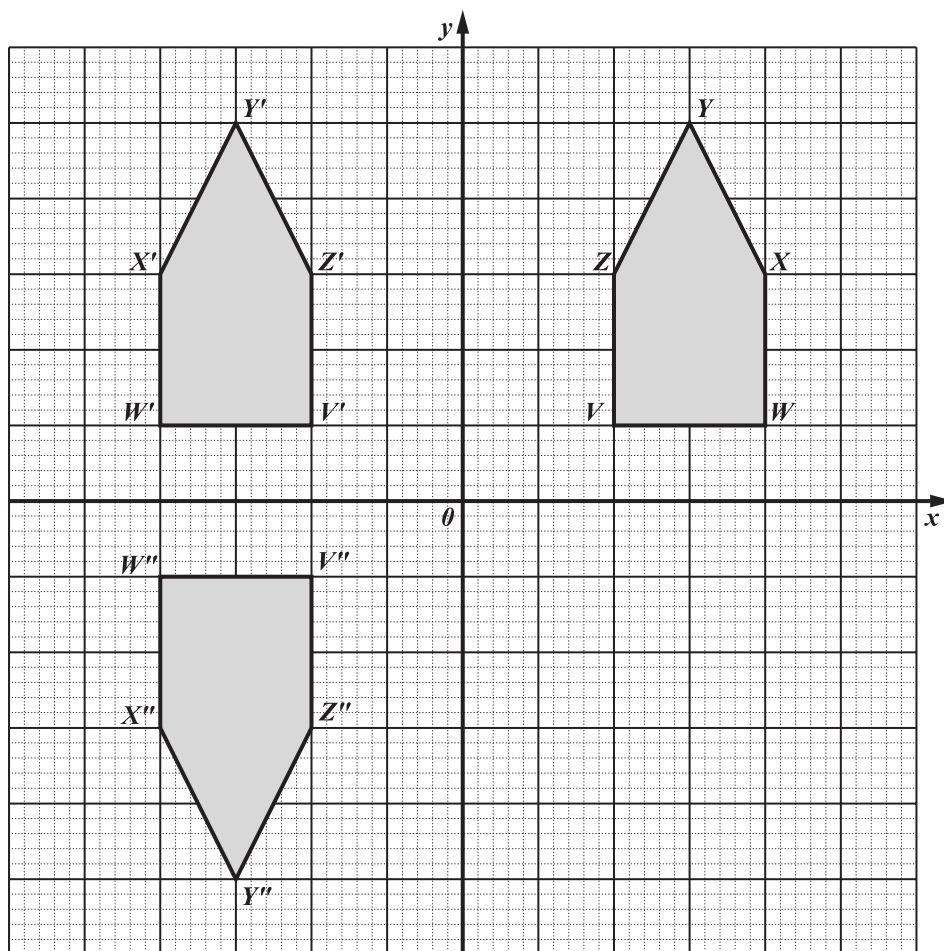


55. The value of Angle FEO is

- (A) 20°
- (B) 28°
- (C) 38°
- (D) 84°

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Item 56 refers to the following diagram which shows a pentagon, $VWXYZ$, and its image $V'W'X'Y'Z'$ and $V''W''X''Y''Z''$, after it undergoes a composite/double transformation.



56. The single transformation that maps Pentagon $VWXYZ$ onto its image $V''W''X''Y''Z''$ is a
- (A) reflection in the line $y = -x$
 - (B) reflection in the y -axis
 - (C) translation with vector $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$
 - (D) rotation of 180° about the origin

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57. The transformation matrix $Q = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ represents
- (A) a 180° rotation about $(0, 2)$
 (B) a reflection in the line $x = 2$
 (C) a reflection in the line $y = 2$
 (D) an enlargement with scale factor 2
58. If $|A| = 0$, then Matrix A is
- (A) an inverse matrix
 (B) a singular matrix
 (C) an identity matrix
 (D) a non-singular matrix
59. Given that P and Q are points with coordinates $P(1, 3)$ and $Q(-1, 4)$, then the position vector PQ is
- (A) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 (B) $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$
 (C) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$
 (D) $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$
60. The 2×2 matrix $L = \begin{bmatrix} 2 & a \\ b & -2 \end{bmatrix}$, where a and b are constants, maps the point $E(2, 5)$ onto the point $F(-1, 2)$.

The value of a and the value of b are

	a	b
(A)	-1	6
(B)	-1	-6
(C)	1	-6
(D)	1	6

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE

1. Fill in all the information requested clearly in capital letters.

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SUBJECT MATHEMATICS – Paper 01

PROFICIENCY GENERAL

REGISTRATION NUMBER

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FULL NAME _____
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2. Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.
3. Keep it in a safe place until you have received your results.

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Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

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MATHEMATICS
SPECIMEN PAPER 01
KEYS

Item Number	Topic	Specific Objective	Cognitive Level	Key
1	Number Theory and Computation – Significant Figures	1.1.9	CK	C
2	Number Theory and Computation – Scientific notation	1.1.12	AK	C
3	Number Theory and Computation – Evaluating numerical expressions	1.1.3	AK	D
4	Number Theory and Computation – Problem Solving	1.1.11/19	R	A
5	Consumer Arithmetic – Profit & Loss	1.2.1	CK	A
6	Consumer Arithmetic – Simple Interest	1.2.6	AK	A
7	Consumer Arithmetic – Rates	1.2.10	AK	C
8	Consumer Arithmetic – Utilities	1.2.10	R	A
9	Sets – Types of Sets	1.3.1	CK	B
10	Sets – Venn Diagrams	1.3.7	AK	B
11	Sets – Number of Elements in Two Intersecting Sets	1.3.4	R	C
12	Measurement – Problem Solving Involving Time	1.4.10	CK	B
13	Measurement – Converting Units, Capacity	1.4.1	AK	A
14	Measurement – Area	1.4.13	AK	B
15	Measurement – Area of Circles	1.4.6	R	B
16	Algebra 1 – Translating Verbal Statements to Symbolic Expressions	1.5.2	CK	C
17	Algebra 1 – Linear Equations	1.5.10	AK	A
18	Algebra 1 – Simplifying Algebraic Fractions	1.5.8	R	D
19	Introduction to Graphs	1.6.2	CK	A
20	Introduction to Graphs	1.6.3	R	B
21	Statistics 1 – Problem Solving	2.1.9/11	CK	D
22	Statistics 1 – Measures of Central Tendency	2.1.4	AK	B
23	Statistics 1 – Simple Probability	2.1.10	AK	D
24	Statistics 1 – Problem Solving, Measures of Central Tendency	2.1.4	R	B
25	Algebra 2 – Factorization	2.2.1	CK	B
26	Algebra 2 – Changing the Subject of Formulas	2.2.2	AK	A
27	Algebra 2 – Direct Variation	2.2.9	R	B
28	Algebra 2 – Problem Solving	2.2.6	R	A
29	Relations, Functions & Graphs – Equation of a straight line	2.3.7	CK	D
30	Relations, Functions & Graphs – Representing Functions	2.3.2	AK	C
31	Relations, Functions & Graphs – Evaluating Composite Functions	2.3.14	AK	A
32	Relations, Functions & Graphs – Equation of Perpendicular Lines	2.3.7/8	R	B
33	Geometry & Trigonometry 1 – Pythagoras’ Theorem	2.4.7	CK	C
34	Geometry & Trigonometry 1 – Trigonometric Ratios	2.4.8	AK	B

35	Geometry & Trigonometry 1 – Bearing, Problem Solving	2.4.9	AK	D
36	Geometry & Trigonometry 1 – Similar Triangles	2.4.6	R	C
37	Vectors & Matrices 1 – Order of a Matrix	2.5.3	CK	C
38	Vectors & Matrices 1 – Matrix Operation	2.5.4	CK	B
39	Vectors & Matrices 1 – Scalar Quantities	2.5.1	AK	A
40	Vectors & Matrices 1 – Vector Algebra, Triangle Law	2.5.2	R	D
41	Statistics 2 – Problem Solving	3.1.9	CK	B
42	Statistics 2 – Median/Other Features from Grouped Data	3.1.2/8	AK	B
43	Statistics 2 – Semi-Interquartile Range	3.1.8	AK	A
44	Statistics 2 – The Use of Contingency Tables	3.1.10	R	B
45	Relations, Functions & Graphs – Graphs of Linear & Non-Linear Functions	3.2.6	CK	C
46	Relations, Functions & Graphs 2 – Speed-time Graphs	3.2.5	CK	A
47	Relations, Functions & Graphs 2 – Speed-time Graphs	3.2.5	AK	B
48	Relations, Functions & Graphs 2 – Linear Inequalities	3.2.2	AK	B
49	Relations, Functions & Graphs 2 – Linear Programming	3.2.3	R	A
50	Relations, Functions & Graphs 2 – Linear Programming	3.2.3	R	D
51	Geometry & Trigonometry 2 – Reflection	3.3.3	CK	B
52	Geometry & Trigonometry 2 – Rotation	3.3.3	AK	A
53	Geometry & Trigonometry 2 – Sine & Cosine Rule	3.3.7	AK	A
54	Geometry & Trigonometry 2 – Bearings	3.3.10	AK	D
55	Geometry & Trigonometry 2 – Circle Theorems	3.3.1	R	C
56	Geometry & Trigonometry 2 – Combination of Transformations	3.3.6	R	D
57	Vectors & Matrices 1 – Transformation Matrices	3.4.8	CK	D
58	Vectors & Matrices 1 – Determinant of a 2×2 Matrix	3.4.5	CK	B
59	Vectors & Matrices 1 – Position Vectors	3.4.1	AK	D
60	Vectors & Matrices 1 – Problem Solving	3.4.9	R	A

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SUBJECT

MATHEMATICS – Paper 02

PROFICIENCY

GENERAL

REGISTRATION NUMBER

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SCHOOL/CENTRE NUMBER

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NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

DATE OF BIRTH

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SPECIMEN 2025



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M A T H E M A T I C S

Paper 02 – General Proficiency

2 hours 40 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE sections: I, II and III.
2. Section I has THREE questions; Section II has THREE questions and Section III has THREE questions.
3. Answer ALL questions, writing your answers in the spaces provided in this booklet.
4. Numerical answers that are non-exact should be given correct to 3 significant figures or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
5. Do NOT write in the margins.
6. All working MUST be clearly shown.
7. **A table of formulae is provided on pages 4 and 5 of this booklet.**
8. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
9. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**
10. **Diagrams in this booklet are NOT drawn to scale, unless otherwise stated.**

Required Examination Materials

Electronic calculator
Geometry set

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LIST OF FORMULAE

Volume of a prism $V = Ah$	A – area of a cross-section h – the perpendicular length
Volume of a cylinder $V = \pi r^2 h$	r – radius of the base h – the perpendicular height
Volume of a right circular cone/right pyramid $V = \frac{1}{3} Ah$	A – area of the base h – the perpendicular height
Curved surface area of a cone $CSA = \pi r l$	r – radius of the base l – the slant height of the cone
Surface area of a sphere $SA = 4\pi r^2$	r – radius of the sphere
Volume of a sphere $V = \frac{4}{3} \pi r^3$	r – radius of the sphere
Circumference of a circle $C = 2\pi r$ $C = \pi d$	r – radius of the circle d – diameter of the circle
Arc length $S = \frac{\theta}{360} \times 2\pi r$	θ – the angle subtended by the arc, measured in degrees
Area of a circle $A = \pi r^2$	r – radius of the circle
Area of a sector $A = \frac{\theta}{360} \times \pi r^2$	θ – the angle of the sector, measured in degrees
Area of a trapezium $A = \frac{1}{2} (a + b) h$	a and b – the lengths of the parallel sides h – the perpendicular distance between the parallel sides
Simple interest $SI = \frac{P \times R \times T}{100}$	P – principal (initial amount) R – annual rate of interest T – time (in years)
Compound interest $A = P \left(1 + \frac{r}{100} \right)^n$	A – total amount after n years P – principal (initial amount) r – annual rate of interest n – number of years money is invested
Depreciation $A = P \left(1 - \frac{r}{100} \right)^n$	A – value of item after depreciation P – initial value of the item r – annual rate of depreciation n – number of years item depreciates

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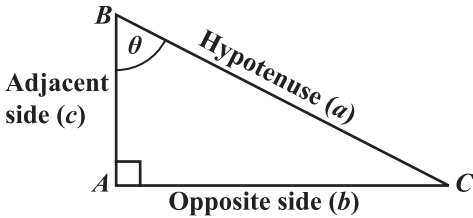
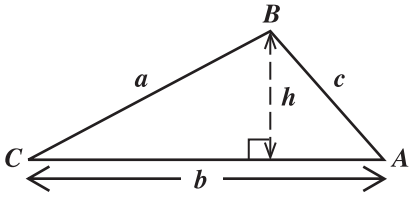
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LIST OF FORMULAE (continued)

Roots of quadratic equations If $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	a – the coefficient of x^2 b – the coefficient of x c – the constant term
Trigonometric ratios $\sin \theta = \frac{b}{a}$ $\cos \theta = \frac{c}{a}$ $\tan \theta = \frac{b}{c}$	 a – length of the hypotenuse b – length of the opposite side c – length of the adjacent side
Pythagoras' theorem $a^2 = b^2 + c^2$	
Area of a triangle $\text{Area of } \Delta = \frac{1}{2} bh$	 b – length of the base of the Δ h – the perpendicular height of the Δ
$\text{Area of } \Delta ABC = \frac{1}{2} ab \sin C$	a and b – the lengths of the adjacent sides of the Δ C – the included angle.
Area of $\Delta ABC =$ $\sqrt{s(s-a)(s-b)(s-c)}$	$s = \frac{a+b+c}{2}$ – the semi-perimeter of the Δ a, b and c – the sides of the Δ
Sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	a, b and c – the sides of the Δ A, B and C – the angles opposite the corresponding sides of the Δ
Cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$	a, b and c – the sides of the Δ A – the angle opposite Side a
Counting formula: union of 2 sets $n(A \cup B) =$ $n(A) + n(B) - n(A \cap B)$	A and B – two finite intersecting sets

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Module 1 – Fundamentals of Secondary-Level Mathematics

Answer ALL questions.

All working MUST be clearly shown.

1. (a) Determine the value of the following, giving your answer in standard form.

$$\frac{(3.29)^2 - 5.5}{\sqrt{1.5 \times 0.06}}$$

.....
(2 marks)

- (b) The diagram below shows rates of exchange at a cambio.

US\$1.00 = TT\$6.45
BBD\$1.00 = TT\$3.00

- (i) Using the table, calculate the BBD equivalent of US\$1.00.

.....
(2 marks)

GO ON TO THE NEXT PAGE

- (ii) Gail exchanged BBD\$1 806.00 for US dollars. Calculate the amount she received in US dollars.

.....
(1 mark)

- (c) John's electricity bill is based on the number of kWh of electricity that he consumes for the month. He is charged \$5.10 per kWh of electricity consumed. The two meter readings for the month of March 2016 are displayed in the diagram below.

01 March	03011	kWh
31 March	03307	kWh

- (i) Calculate the TOTAL amount that John paid for his electricity consumption for the month of March 2016.

.....
(2 marks)

- (ii) For the next month, April 2016, John paid \$2 351.10 for his electricity consumption. Determine his meter reading at the end of April 2016.

.....
(2 marks)

Total 9 marks

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2. Desron forms a sequence using natural numbers, the cubes of these numbers and operations on these numbers and their cubes.

- (a) Firstly, Desron cubed the natural numbers (1, 2, 3, 4, 5, ...) and calculated the sum of the cubes of the first 2, then the first 3, the first 4 and so on. He then put these sums **together** with the cube of the number 1 to form a sequence. The first 5 terms of the sequence are shown in the table below.

n	Series	Sum
1	1^3	$\left[\frac{1}{2}(1+1)\right]^2 = 1$
2	$1^3 + 2^3$	$\left[\frac{2}{2}(1+2)\right]^2 = 9$
3	$1^3 + 2^3 + 3^3$	$\left[\frac{3}{2}(1+3)\right]^2 = 36$
4	$1^3 + 2^3 + 3^3 + 4^3$	$\left[\frac{4}{2}(1+4)\right]^2 = 100$
5	$1^3 + 2^3 + 3^3 + 4^3 + 5^3$	$\left[\frac{5}{2}(1+5)\right]^2 = 225$
.....		$\left[\frac{8}{2}(1+8)\right]^2 = \dots\dots\dots$
$\vdots$	$\vdots$	$\vdots$
n		$\left[\frac{\dots\dots}{2}(1+\dots\dots)\right]^2 = \dots\dots\dots$

Study the pattern of numbers in each row of the table above. Each row relates to one of the terms (cubed numbers) in the sequence of numbers. Some rows have not been included in the table.

Complete the table by filling in the missing information.

(6 marks)

GO ON TO THE NEXT PAGE

- (b) Desron added the first 2 natural numbers, then the first 3, the first 4 and so on. He noted that

$$1 + 2 = 3 = \sqrt{9}$$

$$1 + 2 + 3 = 6 = \sqrt{36}$$

$$1 + 2 + 3 + 4 = 10 = \sqrt{100} .$$

Using the information in the table **on page 8** and the pattern derived from the 3 statements, determine

- (i) the value of x for which $1 + 2 + 3 + 4 + 5 + 6 = \sqrt{x}$

.....
(2 marks)

- (ii) a formula in terms of n for the series $1 + 2 + 3 + 4 + \dots + n$.

.....
(1 mark)

Total 9 marks

3. (a) (i) Factorize completely

$$p^3q^2 + pq^5m.$$

.....
(1 mark)

- (ii) If $a * b = 2a - 5b$, calculate the value of

a) $3 * 4$

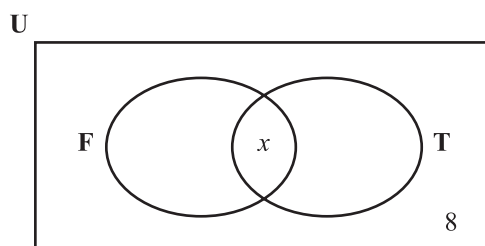
.....
(1 mark)

b) $(3 * 4) * 1.$

.....
(1 mark)

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- (b) The incomplete Venn diagram below shows the number of students in a class of 28 who play football and tennis.



$U = \{\text{all students in the class}\}$
 $F = \{\text{students who play football}\}$
 $T = \{\text{students who play tennis}\}$

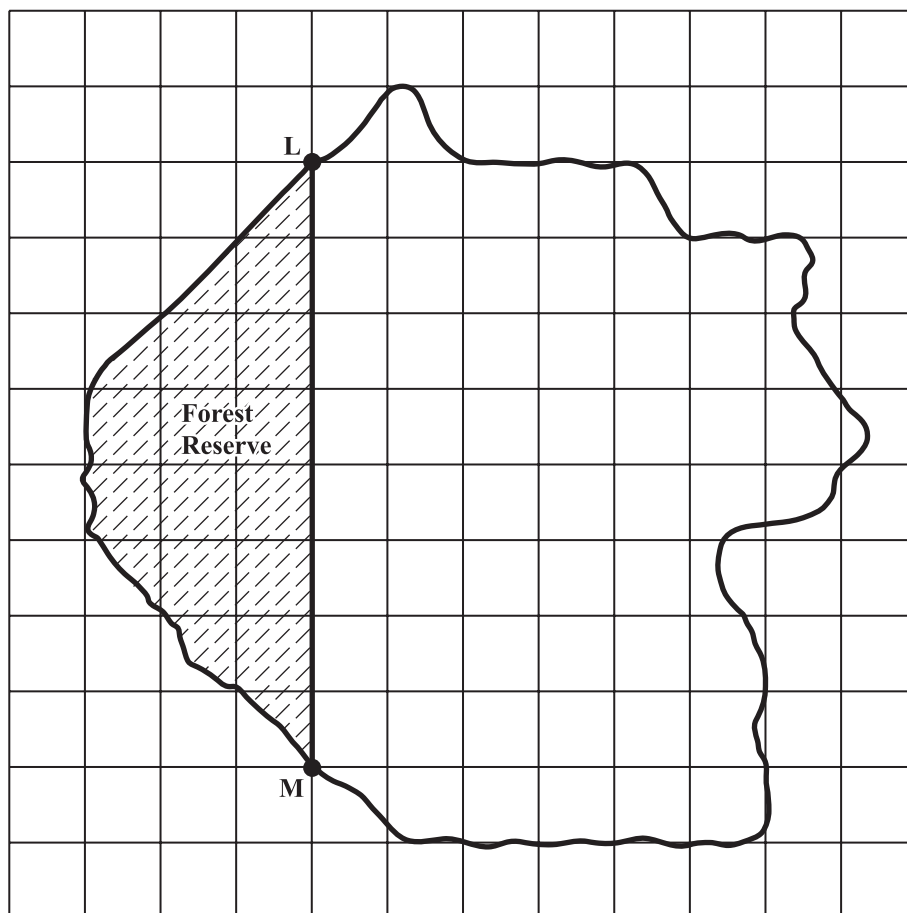
Additional information about the class is that

12 students play tennis
 15 students play football
 8 students play neither football nor tennis
 x students play BOTH football and tennis.

- (i) Complete the Venn diagram above to represent the information, showing the number of students in EACH subset. **(2 marks)**
- (ii) Calculate the value of x .

.....
(2 mark)

- (c) The diagram below shows a map of an island drawn on a grid where each small square measures 1 cm by 1 cm. The map is drawn to a scale of 1:50 000.



- (i) L and M are two tracking stations. State, in cm, the distance LM on the map.

.....
(1 mark)

GO ON TO THE NEXT PAGE

- (ii) The actual distance between two towns on the island is 475 000 cm. What is this actual distance in km?

.....
(1 mark)

- (iii) Calculate the ACTUAL area, in km^2 , of the forest reserve given that

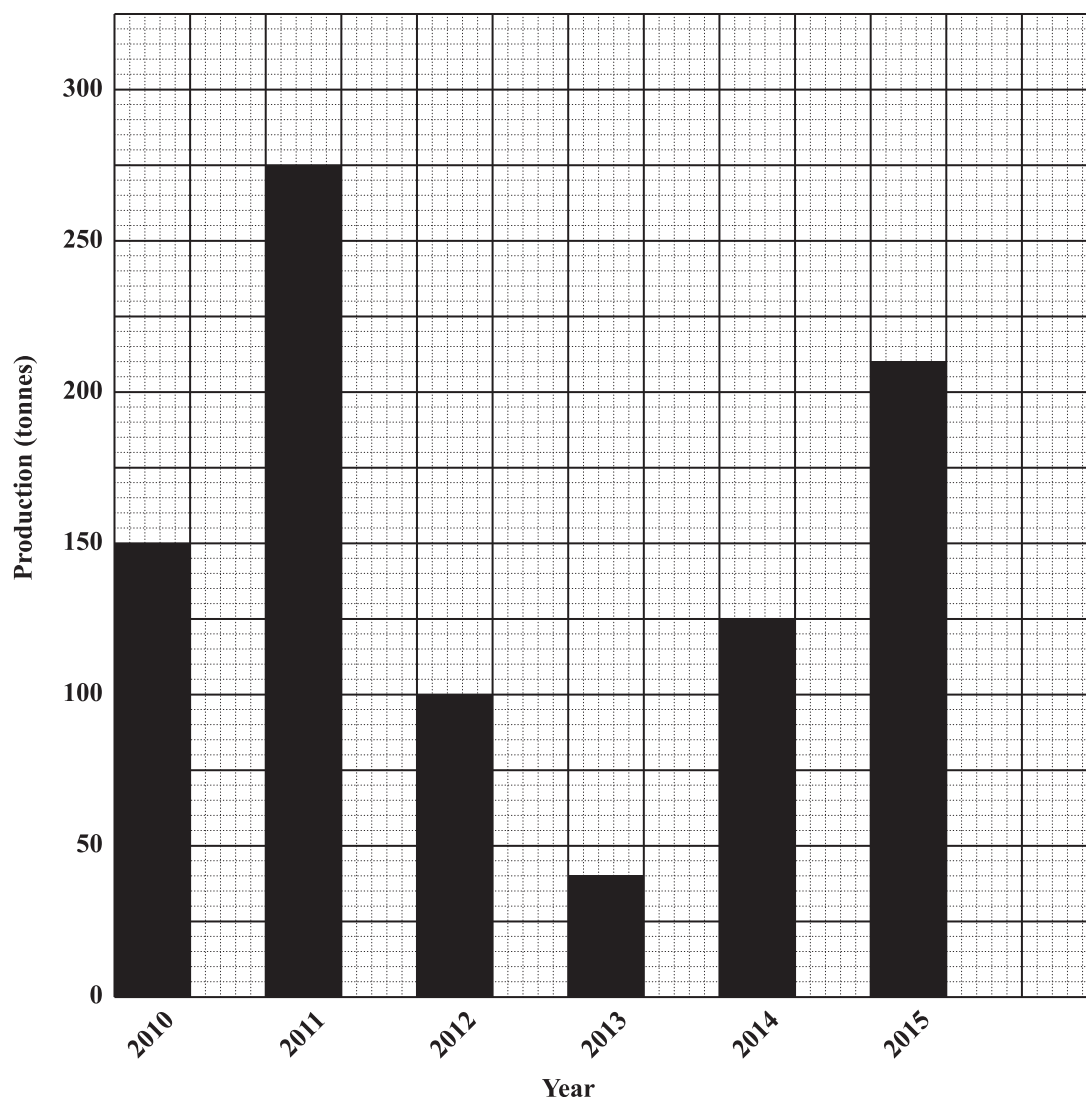
$$1 \times 10^{10} \text{ cm}^2 = 1 \text{ km}^2.$$

.....
(3 marks)

Total 9 marks

Module 2 – Intermediate Secondary-Level Mathematics**Answer ALL questions.****All working MUST be clearly shown.**

4. (a) The bar graph below shows the quantity of bananas (to the nearest tonne) produced annually on a farm over a period of 6 years.



GO ON TO THE NEXT PAGE

- (i) Determine the range of the number of tonnes of bananas produced from 2010 to 2015.

.....
(1 mark)

- (ii) During which year was there the GREATEST production of bananas?

.....
(1 mark)

- (iii) Calculate the mean number of tonnes of bananas produced over the six-year period.

.....
(2 marks)

- (b) In a football tournament, points are awarded as follows: 3 points for a win, 1 point for a draw and 0 points for a loss.

(i) Write a 3×1 matrix, P , to represent this information.

.....
(1 mark)

- (ii) During the tournament, Team Alpha recorded 5 wins, 1 draw and 3 losses, while Team Beta recorded 3 wins, 4 draws and 2 losses.

Write a 2×3 matrix, R , to represent this information.

.....
(1 mark)

- (iii) Calculate the matrix product RP .

.....
(2 marks)

GO ON TO THE NEXT PAGE

- (iv) What does the matrix product RP represent?

.....

.....

.....

.....

(1 mark)

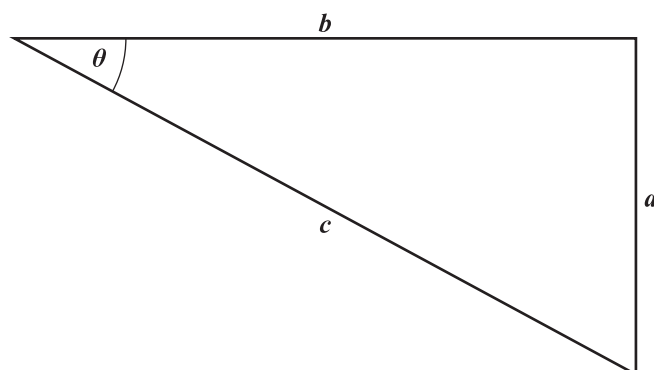
Total 9 marks

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5. (a) Using a ruler, a pencil and a pair of compasses, construct the right-angled triangle ABC , such that $AB = 5$ cm, $\angle ABC = 90^\circ$ and $\angle BAC = 60^\circ$.

.....
(4 marks)

- (b) The diagram below shows a right-angled triangle with sides a units, b units and c units.



Using the diagram,

- (i) express c in terms of a and b .

.....
(1 mark)

- (ii) write, in terms of a , b and c , an expression for $\sin \theta + \cos \theta$.

.....
(2 marks)

- (iii) Using the results from (b) (i) and (ii), show that $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

.....
(2 marks)

Total 9 marks

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6. (a) The line BC passes through the point $A (-5, 3)$ and has a gradient of $\frac{2}{5}$.

(i) The equation of the line BC is of the form $y = mx + c$. Determine the value of c .

.....
(2 marks)

(ii) Determine the equation of the line which passes through the origin and which is perpendicular to the line BC .

.....
(2 marks)

GO ON TO THE NEXT PAGE

- (b) The functions f and g are defined as

$$f(x) = 4x - 5 \quad \text{and} \quad g(x) = \frac{2x - 1}{x + 3}, x \neq -3.$$

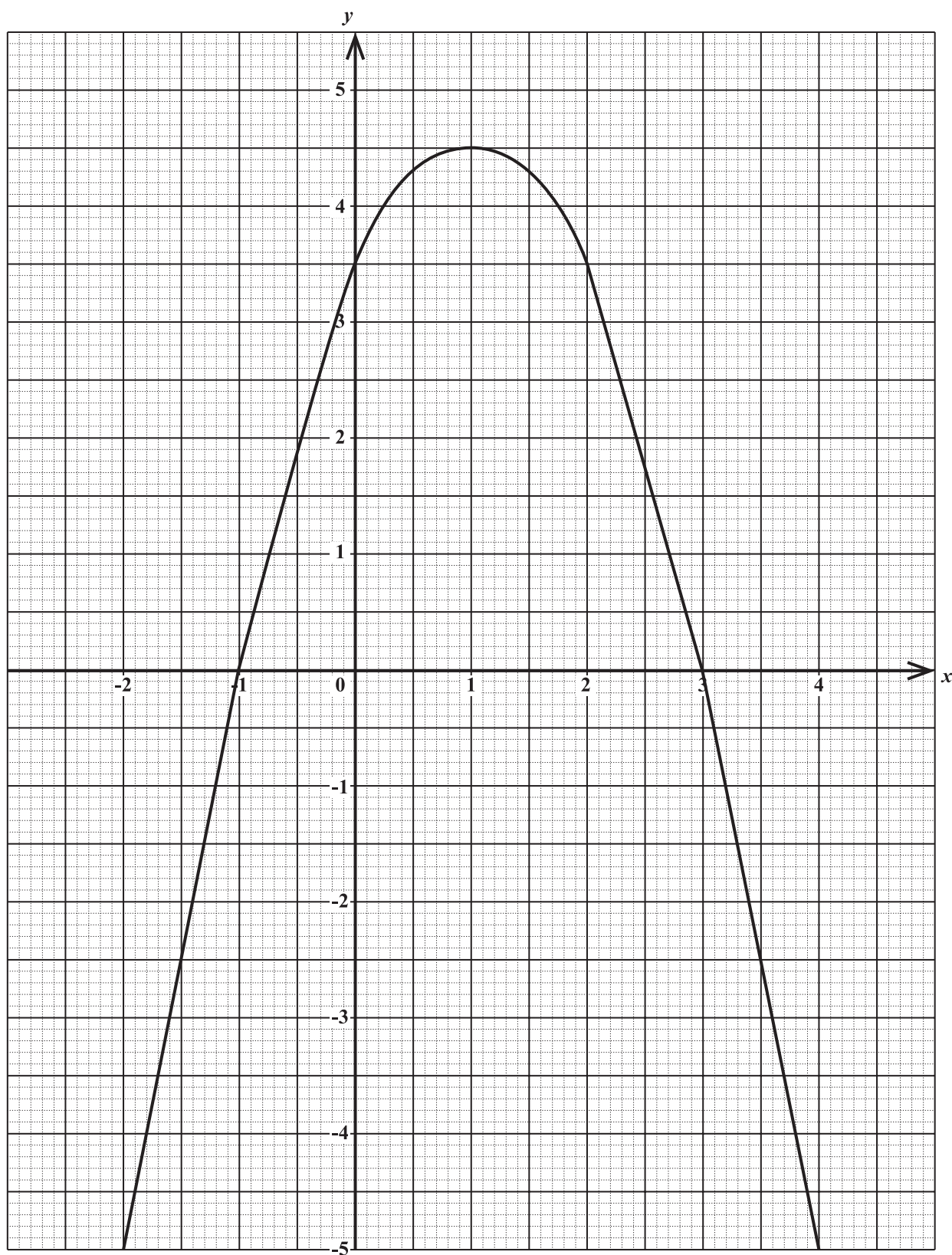
- (i) Determine $gf(3)$.

.....
(2 marks)

- (ii) Derive a simplified expression for $g^{-1}(x)$.

.....
(3 marks)

- (c) The diagram below shows the graph of the quadratic function $y = ax^2 + bx + c$.



GO ON TO THE NEXT PAGE

Using the graph,

- (i) state the coordinates of the turning point of the quadratic function

.....
(1 mark)

- (ii) determine the value of c

.....
(1 mark)

- (iii) write down one of the solutions of the equation $ax^2 + bx + c = 0$.

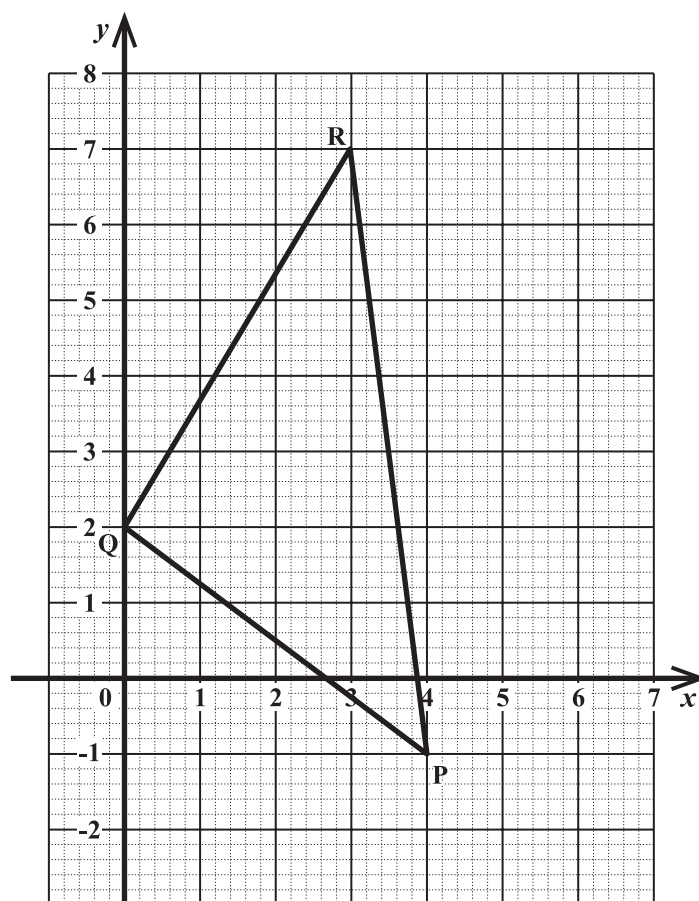
.....
(1 mark)

Total 12 marks

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Module 3 – Higher Concepts in Secondary-Level Mathematics**Answer ALL questions.****All working MUST be clearly shown.**

7. (a) The graph below shows 3 points, P , Q and R , relative to the origin, O .



GO ON TO THE NEXT PAGE

- (i) Write as a column vector in the form $\begin{bmatrix} x \\ y \end{bmatrix}$

a) $\overrightarrow{OP}$

.....
(1 mark)

b) $\overrightarrow{QR}$.

.....
(1 mark)

- (ii) Determine the magnitude of $\overrightarrow{QR}$.

.....
(2 marks)

- (b) Matrices P and Q are such that

$$P = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \text{ and } Q = \begin{pmatrix} 4 & 0 \\ 3 & -1 \end{pmatrix}.$$

- (i) Find P^{-1} , the inverse of P .

.....
(2 marks)

- (ii) Write the following pair of simultaneous equations as a matrix equation.

$$\begin{aligned} 3x + 2y &= 1 \\ 5x + 4y &= 5 \end{aligned}$$

.....
(1 mark)

- (iii) Using a matrix method, solve for x and y .

.....
(2 marks)

Total 9 marks

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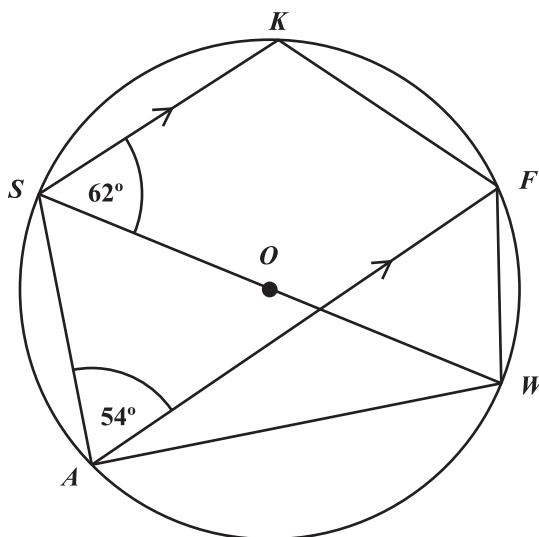
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Barcode Area
Sequential Bar Code

8. (a) In the diagram below, O is the centre of the circle and the points A, S, K, F and W lie on the circumference of the circle.

The line segments SK and AF are parallel with $\angle KSW = 62^\circ$ and $\angle SAF = 54^\circ$.



Calculate, giving reasons for your answer, the measure of

- (i) $\angle FAW$

.....

.....

.....

.....

(2 marks)

- (ii) $\angle SKF$.

.....

.....

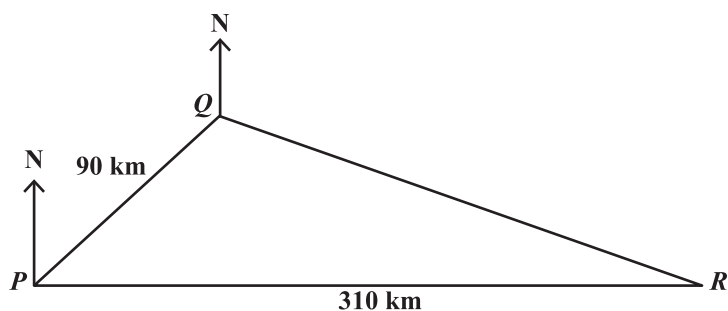
.....

.....

(2 marks)

GO ON TO THE NEXT PAGE

- (b) A ship travels from Penville (P), on a bearing of 030° , to Quintel (Q), 90 km away. It then travels to Rostown (R) which is 310 km due east of Penville (P), as shown in the diagram below.



- (i) Indicate on the diagram the bearing 030° . (1 mark)

- (ii) Determine the bearing of P from Q .

(1 mark)

- (iii) Calculate, to the nearest km, the distance between Q and R .

(3 marks)

Total 9 marks

GO ON TO THE NEXT PAGE

9. (a) The table below shows the amount of time 25 students spent at the school canteen waiting to be served.

Time Spent at the Canteen (minutes)	Frequency	Cumulative Frequency
6–10	2	2
11–15	5	7
16–20	6	13
21–25	8	21
26–30	3	24
31–35	1	25

- (i) On the grid on **page 31**, draw a cumulative frequency curve to show the amount of time the students spent at the canteen. Use a scale of **2 cm to represent 5 minutes** on the horizontal axis and **2 cm to represent 5 students** on the vertical axis.
(5 marks)
- (ii) A student is randomly chosen from the group of students who visited the canteen. Determine the probability that he/she waited more than 20 minutes to be served.

.....
(2 marks)

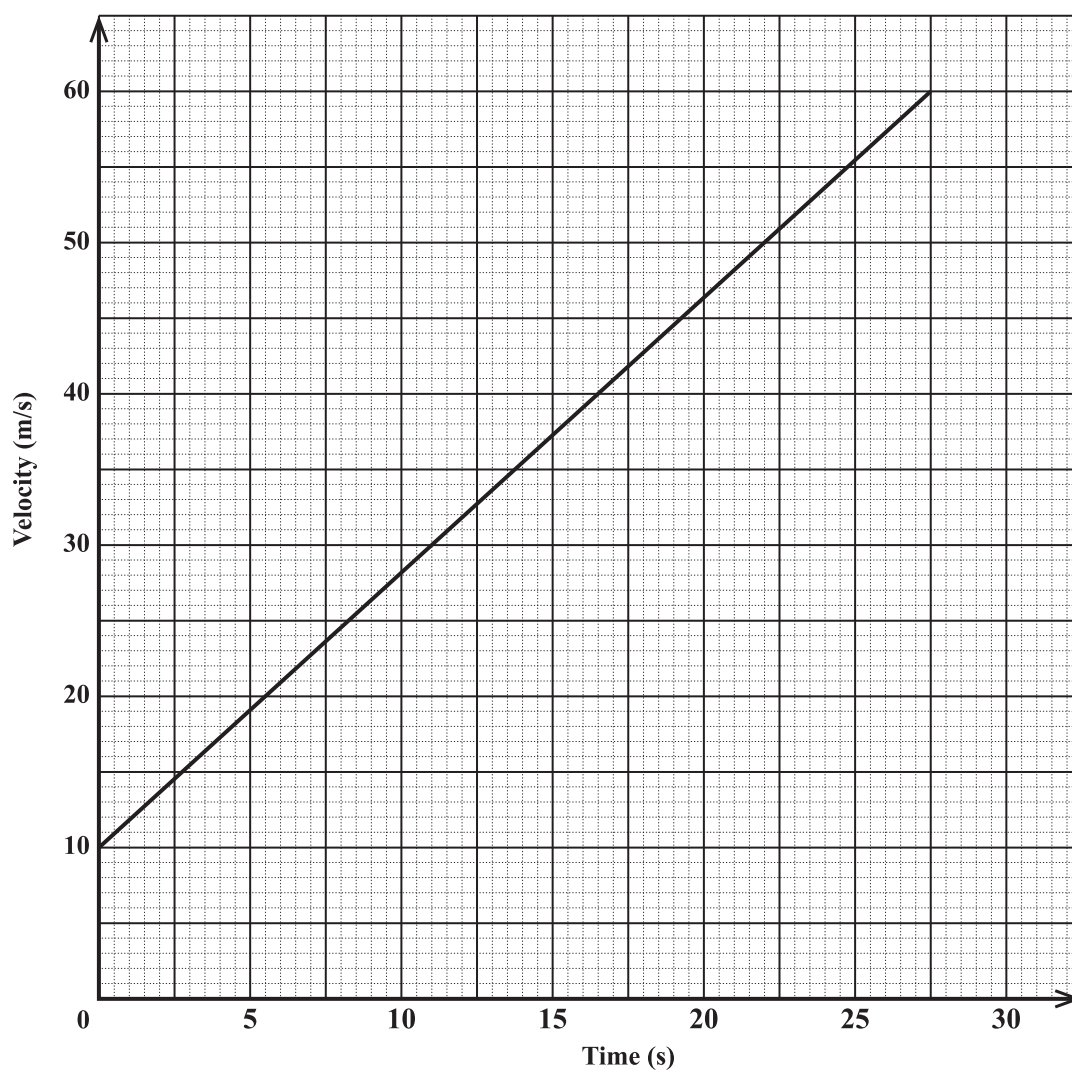
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01234020/CSEC/SPEC 2025

“*”Barcode Area”*”
Sequential Bar Code

- (b) The diagram below represents the velocity of an object over a period of 25 seconds.



GO ON TO THE NEXT PAGE

- (i) Given that the linear motion of the object can be expressed in the form $v = at + u$, where a and u are constants, use the graph to determine the values of a and u .

.....
(3 marks)

- (ii) Calculate the TOTAL distance travelled by the object.

.....
(2 marks)

Total 12 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

01234020/CSEC/SPEC 2025

“*”Barcode Area”
Sequential Bar Code

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

***DO NOT
WRITE ON
THIS PAGE***

CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE

1. Fill in all the information requested clearly in capital letters.

TEST CODE

0	1	2	3	4	0	2	0
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SUBJECT MATHEMATICS – Paper 02

PROFICIENCY GENERAL

REGISTRATION NUMBER

--	--	--	--	--	--	--	--	--	--

FULL NAME _____
(BLOCK LETTERS)

SIGNATURE _____

DATE _____

2. Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.
3. Keep it in a safe place until you have received your results.

INSTRUCTION TO SUPERVISOR/INVIGILATOR

Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

Signature _____
Supervisor/Invigilator

Date _____

01234020/KMS/SPECIMEN 2025

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

MATHEMATICS

PAPER 02 - GENERAL PROFICIENCY

KEY AND MARK SCHEME

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

SECTION I

Module 1 – Fundamentals of Secondary-Level Mathematics

Question 1

Profiles			Total
CK	AK	R	
1		1	
		1	
	1		
1	2	2	5

(a) **Determination of the value of**

$$\frac{(32.9)^2 - 5.5}{\sqrt{1.5} \cdot 0.06}$$

$$= 3589.7 \quad \text{CK}_1 - \text{CAO}$$

$$= 3.5897 \cdot 10^3 \quad \text{R}_1 - \text{in standard form}$$

(b) (i)

Barbados equivalent of US\$1

$$\text{BBD\$1.00} = \text{TT\$3.00}$$

$$\backslash \text{TT\$1.00} = \frac{\text{BBD\$1}}{3} \quad \text{R}_1 - \text{finding BBD equivalent of TT\$1}$$

$$\text{US\$1.00} = \text{TT\$6.45}$$

$$\backslash \text{Barbados equivalent of TT\$6.45}$$

$$= \frac{\cancel{20.000} \cdot 6.45}{\cancel{3.000}} \quad \text{AK}_1 - \text{multiplying BBD equivalent of TT\$1 by 6.45}$$

$$= \$2.15$$

$$\backslash \text{US\$1.00} = \text{BBD\$2.15}$$

(ii) **Gail will receive**

$$\text{BBD\$1806.00} = \text{US\$} \left(\frac{1806}{2.15} \right) \quad \text{AK}_1 - \text{dividing by "her"}$$

$$= \$840.00 \quad \text{BBD equivalent of US\$1}$$

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 1 cont'd

- (c) (i) **Total electricity bill for March 2016**
- $$= (03307 - 03011)' \$5.10 \quad \mathbf{AK_1 - process/method}$$
- $$= 296' \$5.10$$
- $$= \$1509.60 \quad \mathbf{CK_1 - CAO}$$
- (ii) **Meter reading at the end of April**
- $$= 03307 + \left(\frac{\$2351.10}{\$5.10} \right) \quad \mathbf{R_1 - method/process}$$
- $$= 03307 + 461$$
- $$= 03768$$
- $\mathbf{CK_1 - CAO}$

Profiles			Total
CK	AK	R	
1	1		
1		1	
2	1	1	4
3	3	3	9

Specific Objectives: 1.1.3., 1.1.9., 1.1.12., 1.2.9., 1.2.10

Question 2

(a) Completion of table

n	Series	Sum
1	1^3	$\left[\frac{1}{2}(1+1)\right]^2 = 1$
2	$1^3 + 2^3$	$\left[\frac{2}{2}(1+2)\right]^2 = 9$
3	$1^3 + 2^3 + 3^3$	$\left[\frac{3}{2}(1+3)\right]^2 = 36$
4	$1^3 + 2^3 + 3^3 + 4^3$	$\left[\frac{4}{2}(1+4)\right]^2 = 100$
5	$1^3 + 2^3 + 3^3 + 4^3 + 5^3$	$\left[\frac{5}{2}(1+5)\right]^2 = 225$
CK_1 8	AK_1 <u>$1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3$</u>	$\left[\frac{8}{2}(1+8)\right]^2 = \text{R}_1$ <u>1296</u>
N	N	N AK_1R_1
n	CK_1 <u>$1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \dots + n^3$</u>	$\left[\frac{n}{2}(1+n)\right]^2 = \frac{n^2}{4}(1 + n)^2$

Profiles			Total
CK	AK	R	
		1	
	1		
2	2	2	6

Question 2 cont'd

Profiles			Total
CK	AK	R	
1	1	1	3
3	3	3	9

(b) (i) **Value of x**

$$1 + 2 + 3 + 4 + 5 + 6 = \sqrt{x} \quad \text{AK}_1 - \text{method/process}$$

$$\Rightarrow x = (1 + 2 + 3 + 4 + 5 + 6)^2$$

$$x = 21^2$$

$$x = 441 \quad \text{CK}_1 - \text{CAO}$$

(ii) **Formula for nth term**

$$1 + 2 + 3 + 4 + 5 + \dots + n = \sqrt{\frac{n^2}{4}(1+n)^2}$$

$$= \sqrt{\left(\frac{n}{2}\right)^2 (1+n)^2} = \sqrt{\left[\frac{n}{2}(1+n)\right]^2}$$

$$= \frac{n}{2}(1+n)$$

R₁ – correct formula

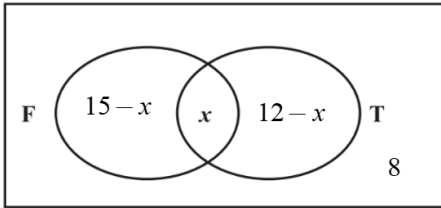
Specific Objectives: 1.1.1., 1.1.2., 1.1.16., 1.2.17

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 3

- (a) (i) **Simplifying**
 $p^3q^2 + pq^5m$
 $= pq^2(p^2 + q^3m)$ **AK₁ – CAO**
- (ii) **a) Finding the value of**
 $a*b = 2a - 5b$
 $3*4 = 2(3) - 5(4)$ **AK₁ – method/process**
 $3*4 = -14$
- b) Finding the value of**
 $(3*4)*1 = 2(-14) - 5(1)$
 $(3*4)*1 = -28 - 5$ **R₁ – both operations correct**
 $(3*4)*1 = -33$
- (b) (i) **Completed Venn diagram**

AK₁ – 15 – x seen
AK₁ – 12 – x seen
- (ii) **Value of x**
 $15 - x + x + 12 - x + 8 = 28$ **R₁ – correct equation**
 $35 - x = 28$
 $x = 35 - 28$
 $x = 7$ **CK₁ – CAO**

Profiles			Total
CK	AK	R	
	1		
	1		
		1	
0	2	1	3
1	2	1	
1	2	1	4
1	4	2	7

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 3 cont'd

- (c) (i)
- Distance LM on map**

$$LM = 8 \text{ cm} \quad \text{CK}_1 - \text{CAO}$$

- (ii)
- Actual distance between the 2 towns on the island**

$$= \frac{475\,000}{100\,000} \text{ km}$$

$$= 4.75 \text{ km} \quad \text{AK}_1 - \text{CAO}$$

- (iii)
- Actual area, in km², of the forest reserve**

Area of forest reserve on map » 15 cm^2

CK₁ – determining the area correctly

Actual area in km²

$$= \frac{15 \times (50\,000)^2}{1 \times 10^{10}} \text{ km}^2 \quad \text{R}_1 - \text{method/process}$$

$$= 3.75 \text{ km}^2 \quad \text{AK}_1 - \text{CAO}$$

Profiles			Total
CK	AK	R	
1			
	1		
1		1	
	1		
2	2	1	5
3	6	3	12
9	12	9	30

Specific Objectives: 1.5.5., 1.5.7., 1.3.6., 1.4.12

TOTAL SECTION I

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

SECTION 2

Module 2 – Intermediate Secondary-Level Mathematics

Question 4

Profiles			Total
CK	AK	R	
1			
	1		
	1		
1			
2	2	0	4

- (a) (i) **The range of the number of tonnes of bananas produced between 2010 and 2015**
- Range = largest to smallest
 Range = 275 - 40
 = 235 CK₁ – CAO
- (ii) **Year in which the production of bananas was the greatest**
- 2011 AK₁ – computation/method/process
- (iii) **Mean**
- AK₁ – computation
/method
/process
- $$\text{Mean} = \frac{150 + 278 + 100 + 40 + 125 + 210}{6}$$
- $$\text{Mean} = \frac{900}{6}$$
- Mean = 150 tonnes CK₁ – CAO

MATHEMATICS

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KEY AND MARK SCHEME

Question 4 cont'd

Profiles			Total
CK	AK	R	
1			
	1		
	1		
1			
		1	
2	2	1	5
4	4	1	9

(b) (i) **Matrix P**

$$P = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \quad \text{CK}_1 - \text{CAO}$$

(ii) **Matrix R**

$$R = \begin{pmatrix} 5 & 1 & 3 \\ 3 & 4 & 2 \end{pmatrix} \quad \text{AK}_1 - \text{correct rows and columns}$$

(iii) **Matrix product, RP**

$$RP = \begin{pmatrix} 5 & 1 & 3 \\ 3 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \quad \text{AK}_1 - \text{computation/method/process}$$

$$RP = \begin{pmatrix} 16 \\ 13 \end{pmatrix} \quad \text{CK}_1 - \text{CAO}$$

(iv) **Representation of Matrix product, RP**

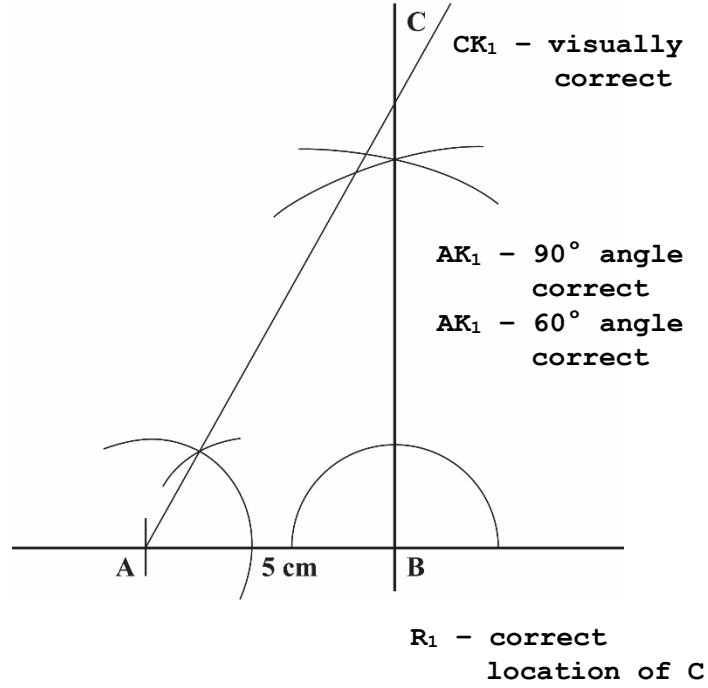
- RP represents the total points for teams Alpha (16) and Beta (13) **R**₁ - **description**

Specific Objectives: 2.1.7., 2.5.4., 2.5.5

Question 5

(a)

Construction of right-angled triangle, ABC



Profiles			Total
CK	AK	R	
1	2	1	

MATHEMATICS

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KEY AND MARK SCHEME

Question 5

Profiles			Total
CK	AK	R	
		1	
		1	
	1		
		1	
	1		
0	2	3	5
1	4	4	9

(b) (i) **Expression of c in terms of a and b**

- From Pythagoras' theorem

$$c^2 = a^2 + b^2 \quad \text{R}_1 - \text{applying Pythagoras' theorem}$$

$$\therefore c = \sqrt{a^2 + b^2}$$

(ii) **Expression for $\sin \theta + \cos \theta$**

$$\sin \theta + \cos \theta = \frac{a}{c} + \frac{b}{c} \quad \text{R}_1 - \text{substitution of ratios}$$

$$\sin \theta + \cos \theta = \frac{a+b}{c}$$

$$\text{but } c = \sqrt{a^2 + b^2} \quad \text{AK}_1 - \text{method/process, sub. } c$$

$$\therefore \sin \theta + \cos \theta = \frac{a+b}{\sqrt{a^2 + b^2}}$$

(iii) **Proof $(\sin \theta)^2 + (\cos \theta)^2 = 1$**

$$(\sin \theta)^2 + (\cos \theta)^2 = \left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 \quad \text{R}_1 - \text{substituting ratios}$$

$$(\sin \theta)^2 + (\cos \theta)^2 = \frac{a^2 + b^2}{c^2}$$

$$\text{but } c^2 = a^2 + b^2$$

$$\text{AK}_1 - \text{method/process, sub. } c^2$$

$$\therefore (\sin \theta)^2 + (\cos \theta)^2 = \frac{c^2}{c^2} = 1$$

Specific Objectives: 2.4.4., 2.4.8., 2.4.10

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 6

Profiles			Total
CK	AK	R	
1	1		
1	1		
2	2	0	4

(a) (i)

Equation of BCGiven $A(-5, 3)$ and $m = \frac{2}{5}$

$$y = mx + c$$

$$3 = \frac{2}{5}(-5) + c$$

$$c = 3 + 2 = 5 \quad \text{AK}_1 - \text{finding } c$$

$$\therefore y = \frac{2}{5}x + 5 \quad \text{CK}_1 - \text{correct equation}$$

OR

$$y - 3 = \frac{2}{5}(x - (-5))$$

$$y - 3 = \frac{2}{5}x + 2 \quad \text{AK}_1 - \text{computation/method/process}$$

$$\therefore y = \frac{2}{5}x + 5 \quad \text{CK}_1$$

(ii)

Equation of line which passes through the origin and is perpendicular to BCUsing $(0, 0)$ and $m = -\frac{5}{2}$ **AK**₁ – correct gradient

$$y = -\frac{5}{2}x \quad \text{CK}_1 - \text{correct equation}$$

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 6 cont'd

Profiles			Total
CK	AK	R	
1	1		
1	1	1	5

(b) (i) **Evaluating composite function gf**

$$f(x) = 4x - 5; \quad g(x) = \frac{2x - 1}{x + 3}$$

$$f(3) = 4(3) - 5 = 7$$

$$gf(3) = g(7) \quad \text{AK}_1 - \text{method/process}$$

$$gf(7) = \frac{2(7) - 1}{7 + 3} = 1.3 \quad \text{CK}_1 - \text{CAO}$$

(ii) **Inverse function $g^{-1}(x)$**

$$g(x) = \frac{2x - 1}{x + 3}, \quad x^1 - 3$$

$$\text{Let } y = g(x)$$

$$\backslash \quad y = \frac{2x - 1}{x + 3}$$

Interchanging y with x yields

$$x = \frac{2y - 1}{y + 3}$$

$$x(y + 3) = 2y - 1 \quad \text{AK}_1 - \text{method/process}$$

$$xy + 3x = 2y - 1$$

$$2y - xy = 1 + 3x$$

$$y(2 - x) = 1 + 3x \quad \text{CK}_1 - \text{collecting like terms}$$

$$y = \frac{1 + 3x}{2 - x}$$

$$\backslash \quad g^{-1}(x) = \frac{1 + 3x}{2 - x} \quad \text{R}_1 - \text{correct inverse}$$

MATHEMATICS

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KEY AND MARK SCHEME

Question 6 cont'd

- (c) (i) **Coordinates of turning point**
 - **(1, 4) R₁ – correct coordinates given**
- (ii) **Value of c**
 According to the graph,
c = 3 R₁ – correct value
- (iii) **Solutions of $ax^2 + bx + c = 0$**
 - **x = -1 or x = 3**
R₁ – either of the solutions correct

Profiles			Total
CK	AK	R	
		1	
		1	
		1	
0	0	3	3
4	4	4	12

Specific Objectives: 2.3.7., 2.3.8., 2.3.11., 2.3.13., 2.3.16., 2.3.17

TOTAL SECTION II

9	12	9	30
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MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

SECTION 3

Module 3 – Higher Concepts in Secondary-Level Mathematics

Question 7

Profiles			Total
CK	AK	R	
1			
	1		
	1		
1			
2	2	0	4

(a) (i) a) Vector $\overrightarrow{OP}$

$$\overrightarrow{OP} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \quad \text{CK}_1 - \text{CAO}$$

b) Vector $\overrightarrow{QR}$

$$\overrightarrow{QR} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \quad \text{AK}_1 - \text{CAO}$$

(ii) Magnitude of $\overrightarrow{QR}$

$$Q(0,2), R(3,7)$$

$$|\overrightarrow{QR}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$|\overrightarrow{QR}| = \sqrt{(3 - 0)^2 + (7 - 2)^2}$$

$$|\overrightarrow{QR}| = \sqrt{9 + 25} = \sqrt{34} \quad \text{AK}_1 \quad \text{computation/method}$$

$$|\overrightarrow{QR}| = 5.83 \text{ sq.units} \quad \text{/process}$$

OR

$$|\overrightarrow{QR}| = \sqrt{x^2 + y^2} \quad \text{CK}_1 - \text{CAO}$$

$$|\overrightarrow{QR}| = \sqrt{3^2 + 5^2}$$

$$|\overrightarrow{QR}| = \sqrt{9 + 25} = \sqrt{34}$$

$$|\overrightarrow{QR}| = 5.83 \text{ sq.units}$$

Question 7 cont'd

(b) (i) **Inverse of Matrix P**

$$P = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix}$$

$$P^{-1} = \frac{1}{|P|} \times P \text{ Adjoint}$$

$$P^{-1} = \frac{1}{(3 \times 4) - (5 \times 2)} \times \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix}$$

$$P^{-1} = \frac{1}{2} \times \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix}$$

$$P^{-1} = \begin{pmatrix} 2 & -1 \\ -\frac{5}{2} & \frac{3}{2} \end{pmatrix}$$

CK₁ - CAO

(ii) **Matrix equation**

$$\begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}_{\mathbf{R}_1} \text{ - matrices in correct order}$$

(iii) **Solution using matrix equation**

$$\begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

Pre-multiplying by P^{-1}

$$\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} x & y \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

AK₁ = pre-multiplying

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -6 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

$$\therefore x = -3, y = 5$$

R_1 - equating corresponding elements

Profiles			Total
CK	AK	R	
1	1	1	
1	1	1	
1	2	2	5
3	4	2	9

Specific Objectives:3.4.1., 3.4.2., 3.4.7.,3.4.9.

MATHEMATICS

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KEY AND MARK SCHEME

Question 8.

Profiles			Total
CK	AK	R	
	1	1	
	1	1	
0	2	2	4

(a) (i) **Angle FAW**

$$\angle FAW = 90^\circ - 54^\circ \quad \text{AK}_1 - \text{subtracting from } 90^\circ$$

$$\angle FAW = 36^\circ$$

SW is a diameter of the circle

hence, $\angle SAW$ is right-angled.**R₁ – Reason**(ii) **Angle SKF**

$$\angle SKF = 180^\circ - 54^\circ$$

$$\angle SKF = 126^\circ$$

AK₁ – computation/method/processASKF is a cyclic quadrilateral, **R₁ – Reason** $\therefore \angle SKF$ and $\angle SAF$ are supplementary.

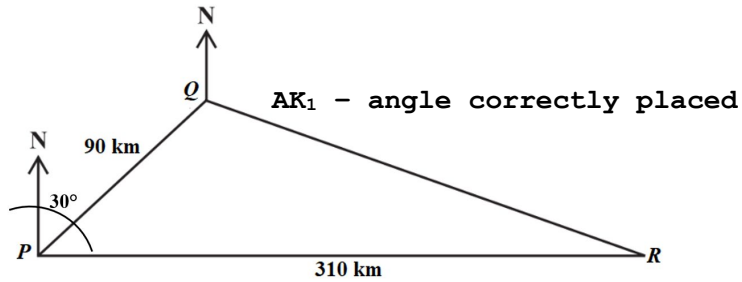
MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 8

Profiles			Total
CK	AK	R	
	1		
		1	
1	1	1	
1	2	2	5
1	4	4	9

(b) (i) **Completion of diagram**(ii) **Bearing of P from Q**

$$= 360^\circ - [180^\circ - 30^\circ]$$

$$= 360^\circ - 150^\circ$$

$$= 210^\circ$$

R₁ - subtracting from both 180° and 360°

(iii) **Distance between Q and R**

Applying the cosine rule

$$QR^2 = q^2 + r^2 - 2qr \cos p$$

R₁ - applying in the cosine rule

$$QR^2 = 310^2 + 90^2 - 2(310 \times 90) \cos 60^\circ$$

$$QR^2 = 96100 + 8100 - 27900$$

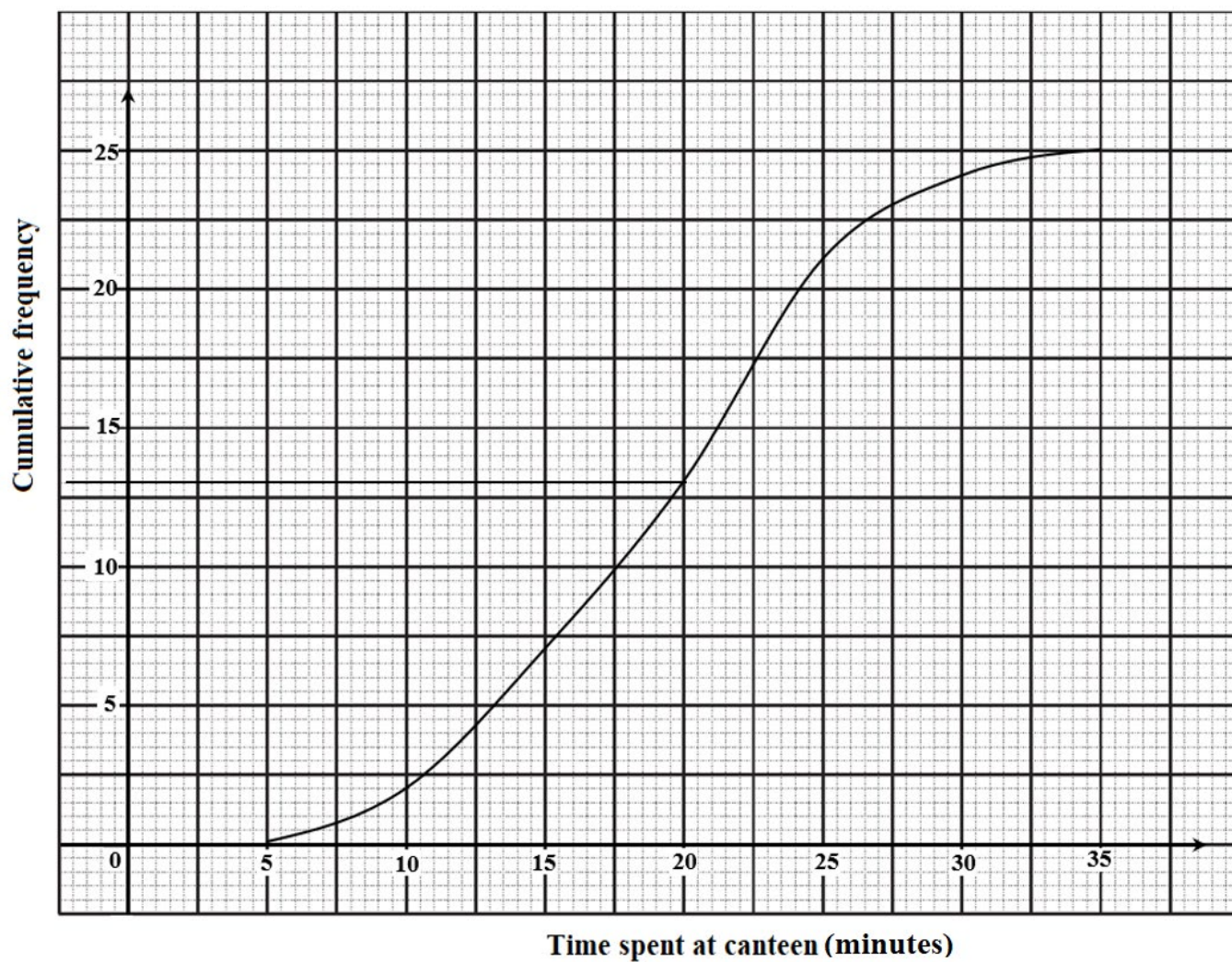
$$QR = \sqrt{76300}$$

AK₁ - method/computation

$$QR = 276.22 \text{ km}$$

CK₁ - CAO

Specific Objectives: 3.3.1., 3.3.7., 3.3.10

Question 9(a)(i)

MATHEMATICS

PAPER 02 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

Question 9 cont'd

(a) (i) Cumulative frequency curve

Labels – both labels correct, 1 mark

CK₁

Plotted points – every 3 points plotted correctly, 1 mark

AK₂

Scale – correct scales used, 1 mark

R₁

Smooth curve – 1 mark

CK₁(ii) Probability: AK₁ – computation/method/processR₁ – correct numerator

$$P(> 20 \text{ minutes}) = \frac{25 - 13}{25}$$

$$P(> 20 \text{ minutes}) = \frac{12}{25} \quad \text{CK}_1 - \text{CAO}$$

$$P(> 20 \text{ minutes}) = 0.68$$

Profiles			Total
CK	AK	R	
2	2	1	
		1	
1			
3	2	2	7

MATHEMATICS

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KEY AND MARK SCHEME

Question 9 cont'd

			Profiles			Total
			CK	AK	R	
(b)	(i)	Values of a and u $a = \text{gradient}$ From the graph, using the points (25, 60) and (0, 10), $a = \frac{60 - 10}{25 - 0}$ R₁ – relating a to the gradient $a = \frac{50}{25}$ $a = 2$ AK₁ – correct gradient $u = \text{the y-intercept}$ $\backslash u = 10$ CK₁ – correct y-intercept	1	1	1	
	(ii)	Total distance travelled Total distance travelled = Area under graph $\backslash d = \frac{1}{2} \times \text{base} \times \text{height}$ $d = \frac{1 \times 25 \times 50}{2}$ AK₁ – correct substitution /method $d = 625 \text{ sq. metres}$ CK₁ – CAO	1	1		
			2	2	1	
			5	4	3	
Specific Objectives: 3.1.7., 3.1.9., 3.2.5						
		TOTAL SECTION III	9	12	9	30

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EXAMINATION

FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE

0	1	2	3	4	0	3	2
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SUBJECT

MATHEMATICS – Paper 032

PROFICIENCY

GENERAL

REGISTRATION NUMBER

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SCHOOL/CENTRE NUMBER

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NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

DATE OF BIRTH

D	D	M	M	Y	Y	Y	Y
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SIGNATURE

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Current Bar Code

Barcode Area
Sequential Bar Code

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